Data Governance

WHAT IS THE DIFFERENCE BETWEEN DATA MANAGEMENT AND DATA GOVERNANCE?

WHY WE NEED TO GOVERN OUR DATA?

Data Management Organization

Company evolution recursive model

PEOPLE-DRIVEN

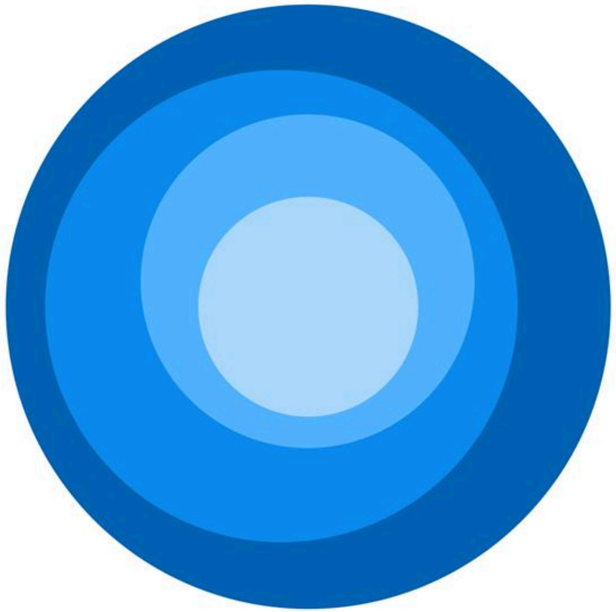
- The company’s activity is based on people’s knowledge and skills.
- Value is created by people.
- Management is task-based.
- Data is not a subject of management.

PROCESS DRIVEN

- Value is created through labor organization technologies.
- Efficiency is the main object of management.
- Discipline is the core element of the culture.
- Automation is a key factor of competitiveness.
- Data is used for historical analysis.

EXECUTION

EFFICIENCY



EFFECTIVENESS

EXCELLENCE

DATA-DRIVEN

- New value is created through data.
- Data delivery speed is a critical factor.
- Data literacy is a key competence.
- At this stage, the organization transitions to a digital operating model.

MODEL-DRIVEN

- Maximum value is achieved through continuously operating digital models.
- Some management functions are delegated to digital algorithms.
- Augmented analytics ensure decision-making speed and excellence.

Data Management Organization

Processes

Data-Driven is a management approach based on data analysis.
The core principle: decisions should be made based on data analysis, not intuition or personal experience.

We strive to make management decisions based on data	Decision-making data is delivered in the form of data products	Data delivery time is a key KPI	The ability to use BI tools is a critical element of a data-driven culture
We aim for all our goals to have measurable characteristics	Every goal should be linked to metrics and KPIs to measure progress and outcomes	Metric Store is a key component of the data platform	The ability to calculate metrics is a key factor in data-driven culture
The quality of decisions depends on the quality and accuracy of data	We must be confident that data can be trusted	Data trust rating	Ensuring trust in data is the main criterion for its business value

Why Manage Data

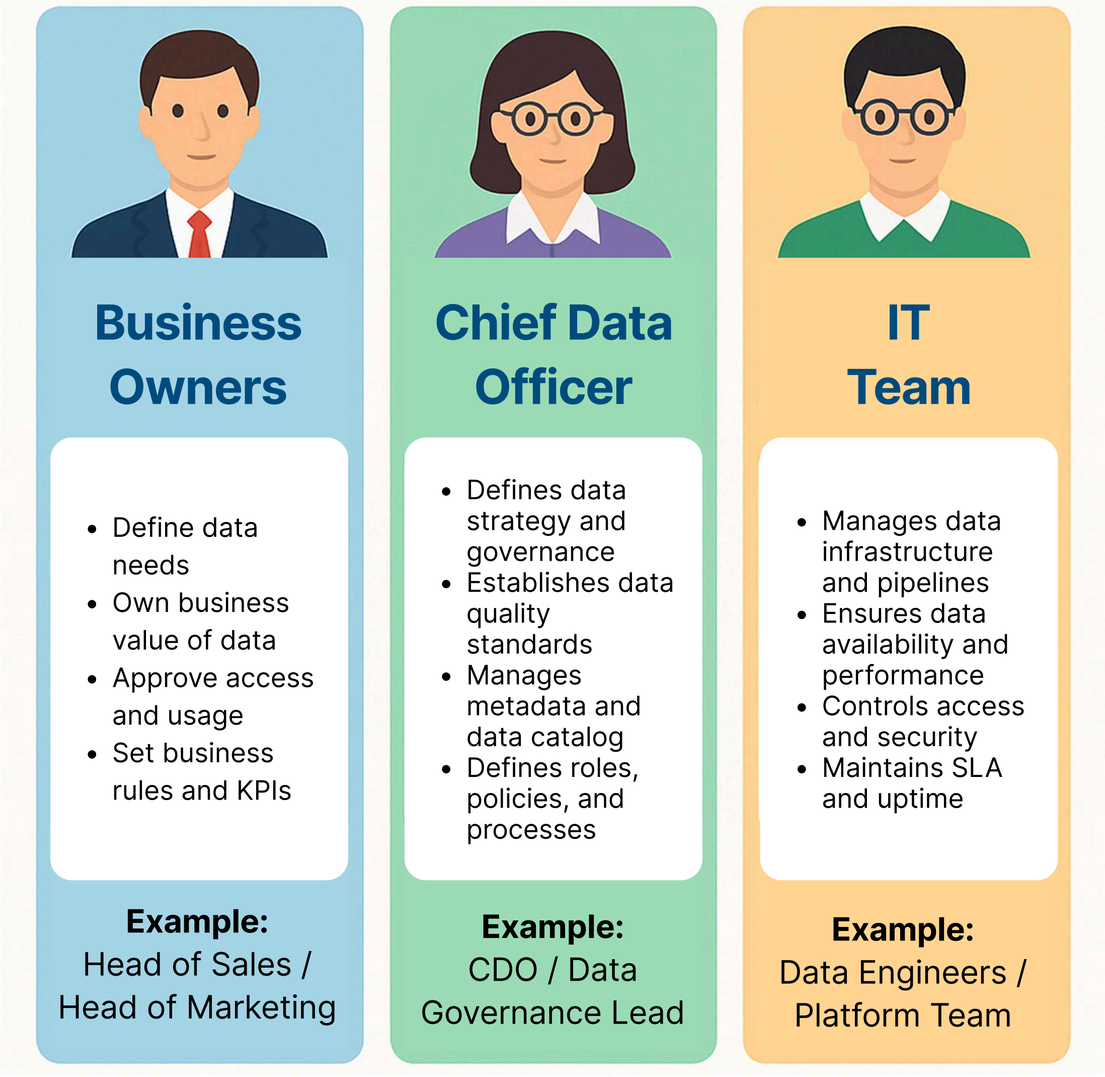
Data Governance

- A set of policies, standards, and procedures.
- Defines who is responsible for data, how it should be used, and what rules apply.
- Key elements:
  - Roles and responsibilities (business owner, CDO/Data steward, IT team)
  - Data policies (quality, security, compliance)
  - Data stewardship — operational support and data quality assurance

Data Governance

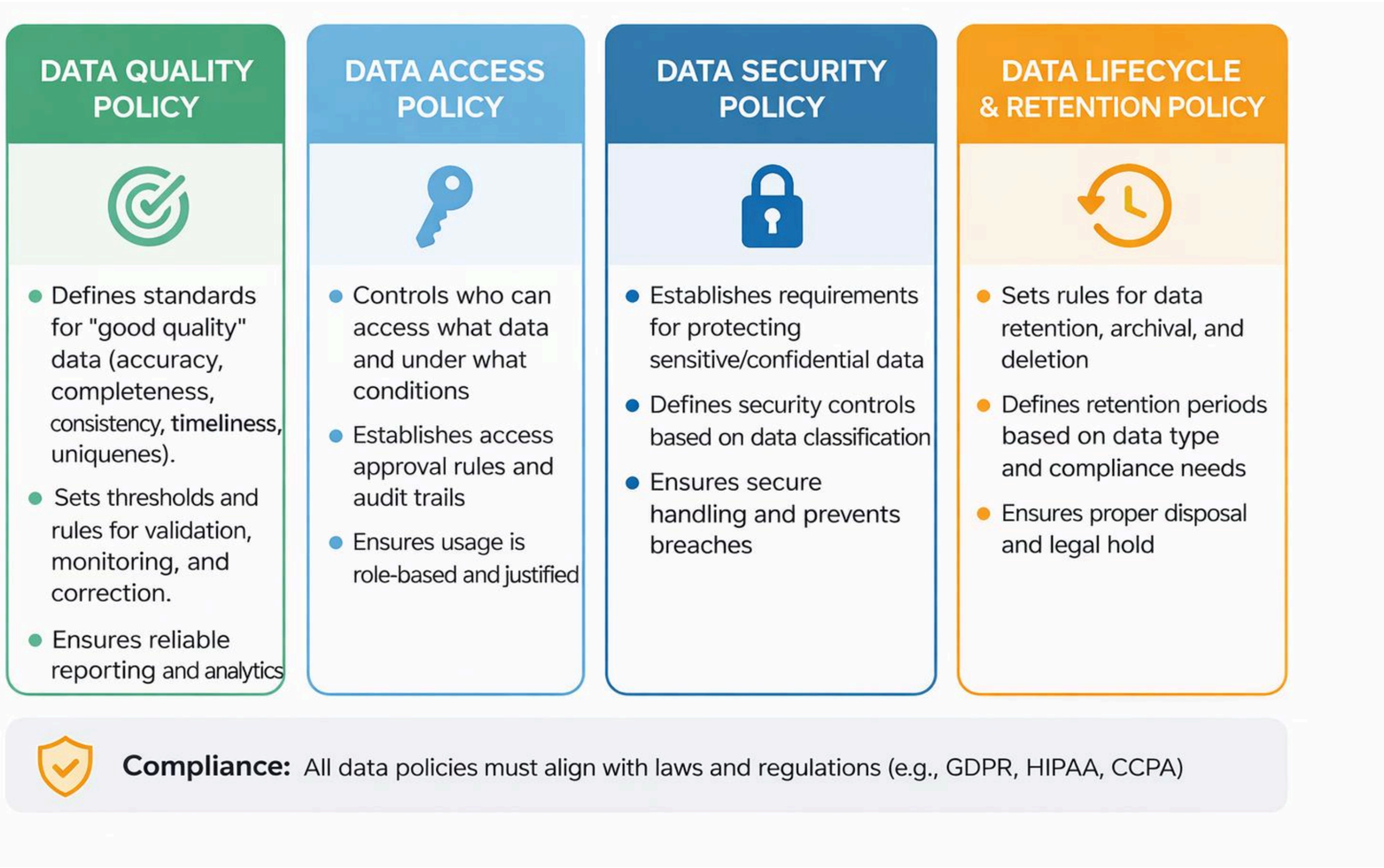
Roles and Responsibilities

Properly defined roles ensure accountability, consistency, and collaboration across the organization.



Data Governance

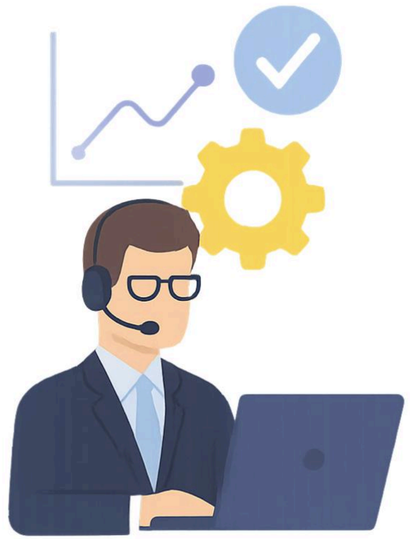
Data Policies



Policies provide the **rules and standards** to manage and use data responsibly

Data Governance

Data Stewardship



DATA STEWARDSHIP

- Monitor and report on data quality KPIs.
- Coordinate across departments to align data definitions and usage.
- Resolve conflicts or ambiguities in data usage.
- Act as a champion for data literacy and responsible use of data.
- Maintain metadata and business glossaries

The ongoing operational role focused on maintaining data quality, integrity, and usability across the organization.

Why Manage Data

Organizational Structure

- How the data management team is organized: centralized, decentralized, or federated model.
- Examples:
  - Centralized — one center of excellence for data.
  - Decentralized — data is managed locally by business units.
  - Federated — a hybrid model: shared standards + local ownership.

Key Components of Data Management Organization

Organizational Structure

Centralized

Consistent governance, easier compliance, centralized expertise

Slower to adapt, possible bottlenecks.

e.g. Small or highly regulated companies (e.g., banking)

All data responsibilities handled by a single central team

Organizational Structure

Decentralized

Flexibility, fast response to local needs.

Data silos, inconsistent definitions, harder alignment.

e.g. Large diversified companies with independent business units.

Each unit manages its own data independently

Organizational Structure

Federated

Balance of control and flexibility, shared responsibility.

Requires strong coordination and clearly defined roles.

e.g. Enterprises with both global and local operations

Central governance with decentralized execution

Organizational Structure

Definition

What is a Domain approach?

A methodology for **organizing data management** in federated organizational structure which responsibility for data is **distributed across business domains**, rather than centralized within an IT team. This approach is especially popular within the **Data Mesh** concept, where each business domain manages its data as a product.

Organizational Structure

Domain approach

Data belongs to the business domain:

- Data is distributed across business domains.
- Each domain is responsible for the quality, availability, and relevance of its data.

Managing data as a product:

- Assigning data owners.
- Clear SLAs (Service Level Agreements) for data availability and quality.
- Data quality metrics to measure and monitor performance.

Decentralized data storage architecture:

- Data marts for BI and analytics.
- Data lakes for storing raw data.
- APIs for real-time data access.
- Support for Data Mesh architecture.

Self-service data access:

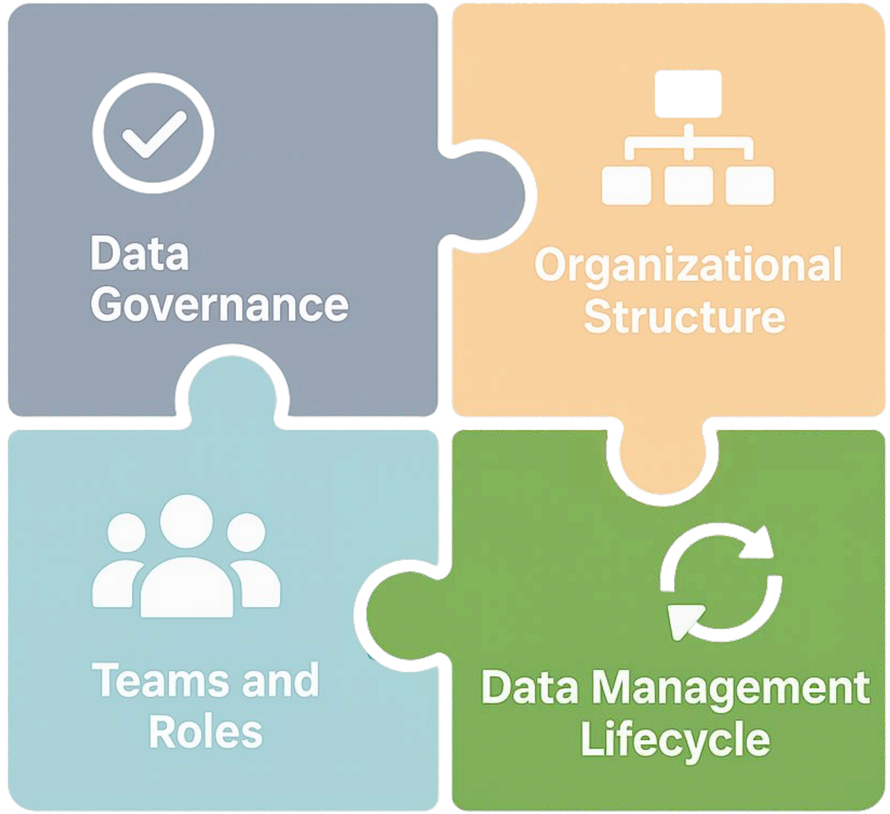
- BI platforms for data visualization.
- Data catalog for data discovery.
- API interfaces for working with data without writing direct SQL queries.

Why Manage Data

Data Management Lifecycle

- How to collect, store and use data

Key Components of Data Management Organization



Data Management Lifecycle

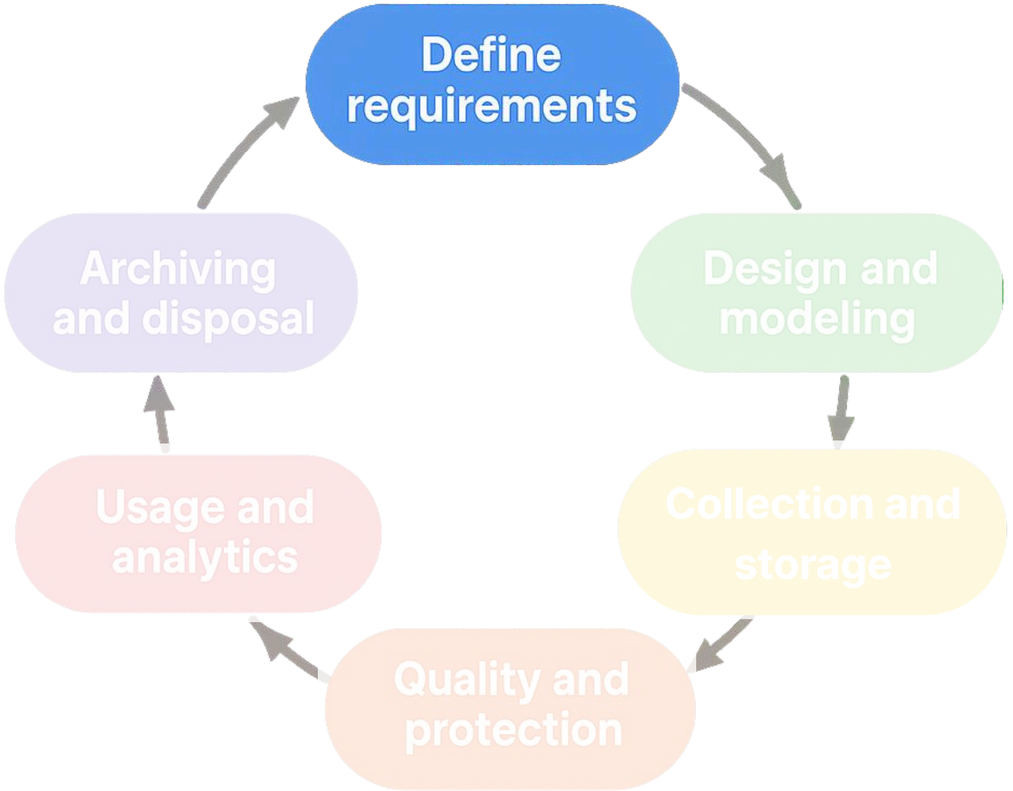
Define Requirements

Goal: Understand what data is needed, for what purposes, and what requirements must be met.

Activities:

- Identifying business needs.
- Defining objectives for data collection (e.g., analytics, reporting, ML).
- Specifying requirements for volume, availability, security, and quality.
- Engaging stakeholders: business units, analysts, IT.

Example: The marketing team wants to collect user behavior data from the website for personalized email campaigns.



Data Management Lifecycle

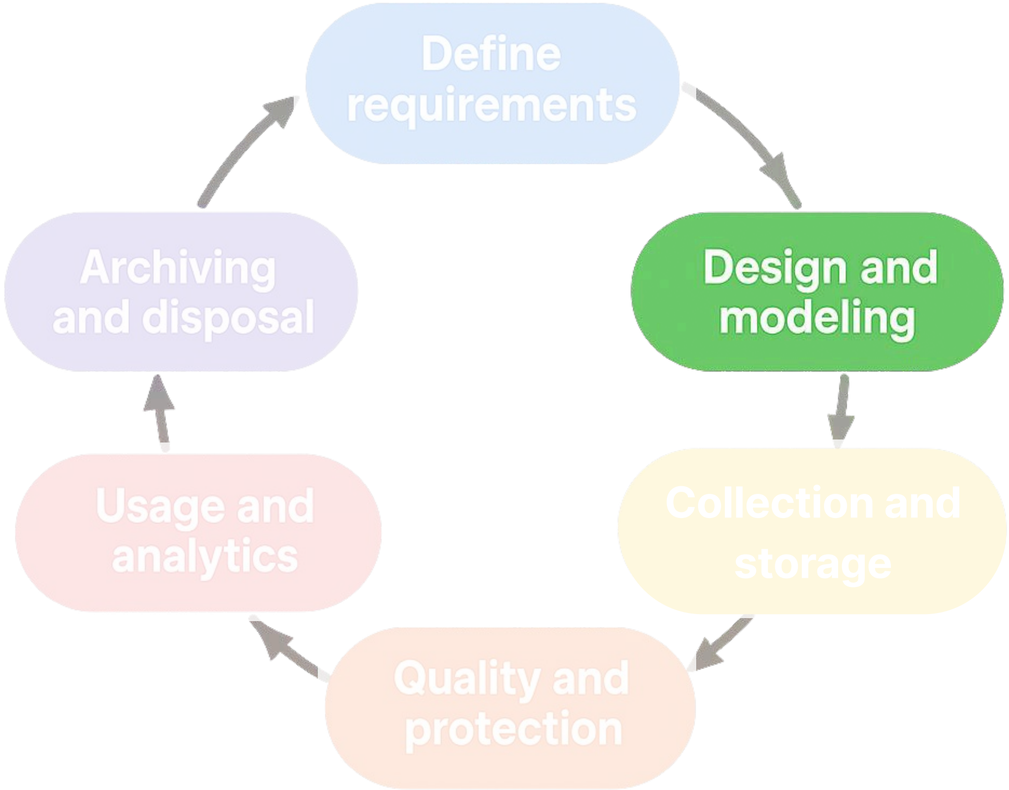
Design and Modeling

Goal: Create the logical and physical structure for data storage and processing.

Activities:

- Designing data models: ER diagrams, Data Vault, Star Schema, etc.
- Planning data flows (ETL/ELT pipelines).Identifying data sources and planning integrations.
- Choosing storage technologies (DWH, lakehouse, databases, etc.)

Designing a “Customer–Purchase–Product” model for the analytical data warehouse



Data Management Lifecycle

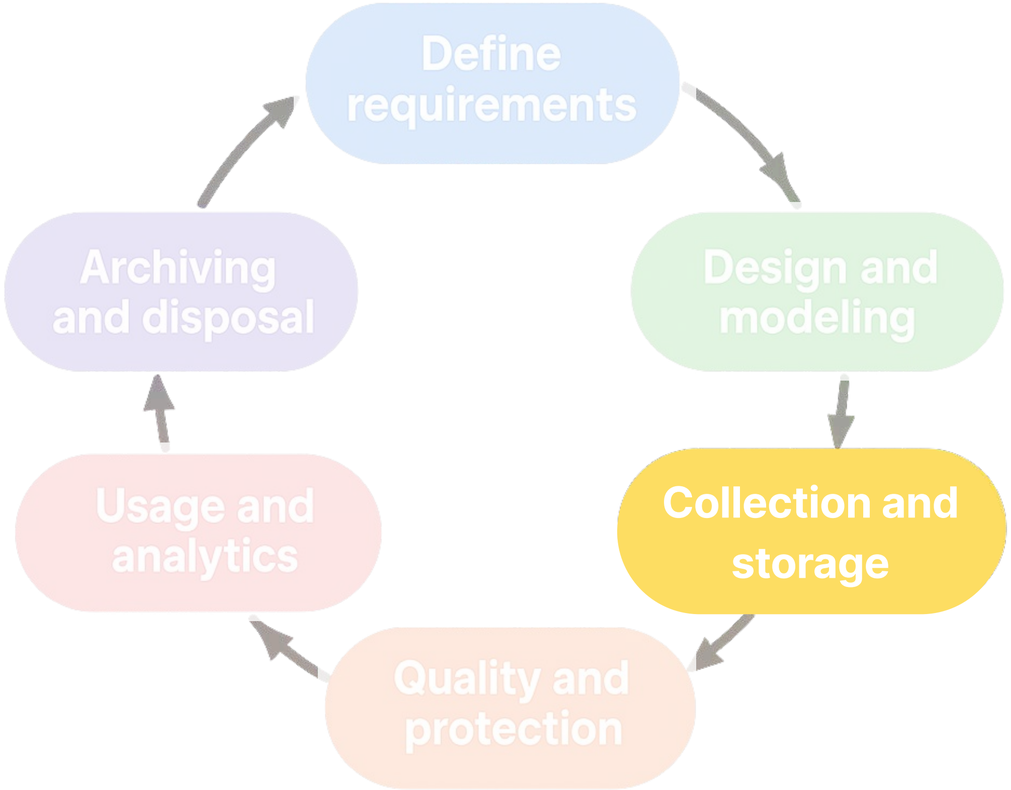
Collection and Storage

Goal: Set up reliable processes for acquiring and storing data.

Activities:

- Connecting to data sources (APIs, databases, files, etc.).
- Ingesting and loading data into the warehouse or data lake.
- Ensuring data durability, backup, and scalability.
- Enriching data using reference or external sources.

Example: Integrating website order data, CRM records, and financial systems into a central repository.



Data Management Lifecycle

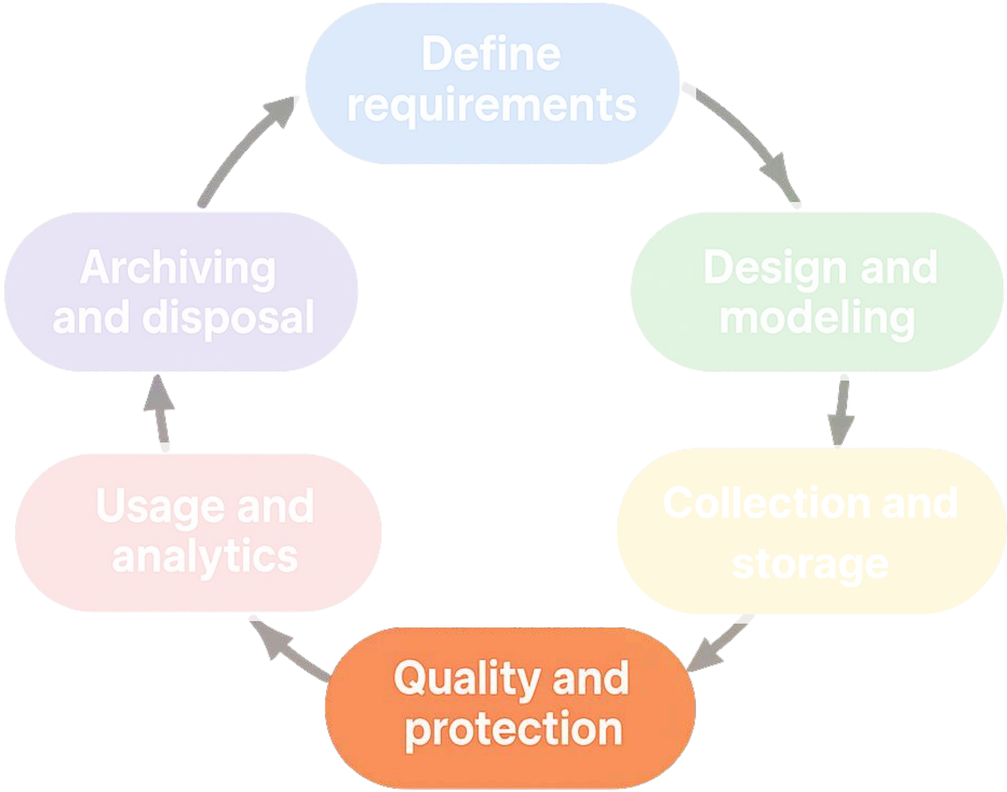
Quality and Protection

Goal: Ensure data is accurate, complete, timely, and secure.

Activities:

- Quality checks: validation, profiling, cleaning, deduplication.
- Real-time data quality monitoring.
- Access control, encryption, and audit logging.
- Compliance with regulations (e.g., GDPR, HIPAA)

Example: Designing a "Customer–Purchase–Product" model for the analytical data warehouse



Data Management Lifecycle

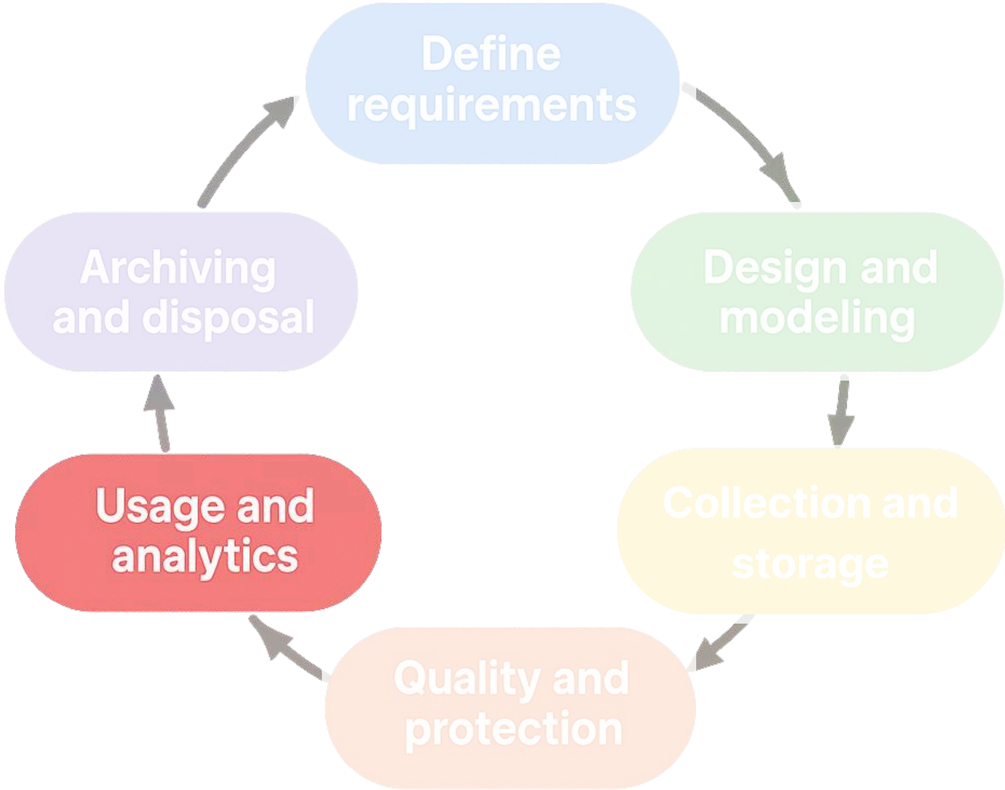
Usage and Analytics

Goal: Provide access to high-quality data for business decision-making.

Activities:

- Creating BI dashboards and reports.
- Making data available for ML models and analytics platforms.
- Managing data versions and trusted sources (“single source of truth”).
- Training users and improving data literacy.

Example: Building a sales performance dashboard for the marketing and sales departments.



Data Management Lifecycle

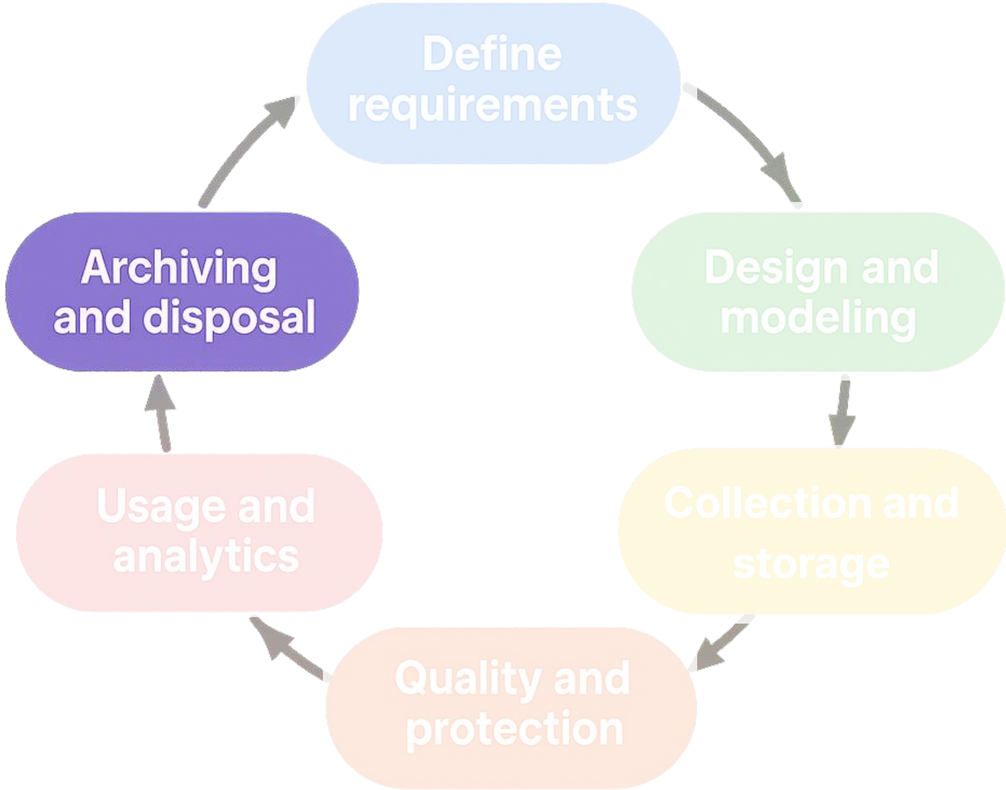
Archiving and Disposal

Goal: Archive or dispose of outdated or unused data according to policy.

Activities:

- Defining retention periods for different data types.
- Automating archiving or deletion processes.
- Protecting against accidental data loss.
- Preparing data for audits or long-term storage.

Example: Archiving order data older than 5 years to cold storage.



Why Manage Data

Teams and Roles

- Chief Data Officer (CDO) — leads the data strategy.
- Data Owner — business accountable for data.
- Data Steward — ensures data quality and metadata accuracy.
- Data Architect / Engineer / Analyst — technical roles.

Key Components of Data Management Organization

Strategic roles

Chief Data Officer (CDO)

Responsibilities:

- Develops and implements the data management strategy.
- Oversees data quality and compliance with standards.
- Ensures data meets regulatory requirements (e.g., GDPR, CCPA).
- Promotes a data-driven culture within the company.

Data Owner

Responsibilities:

- Defines how data is used and who has access to it.
- Ensures data quality and accuracy within their domain.
- Makes decisions about changes to data models.

Operational roles

Data Steward

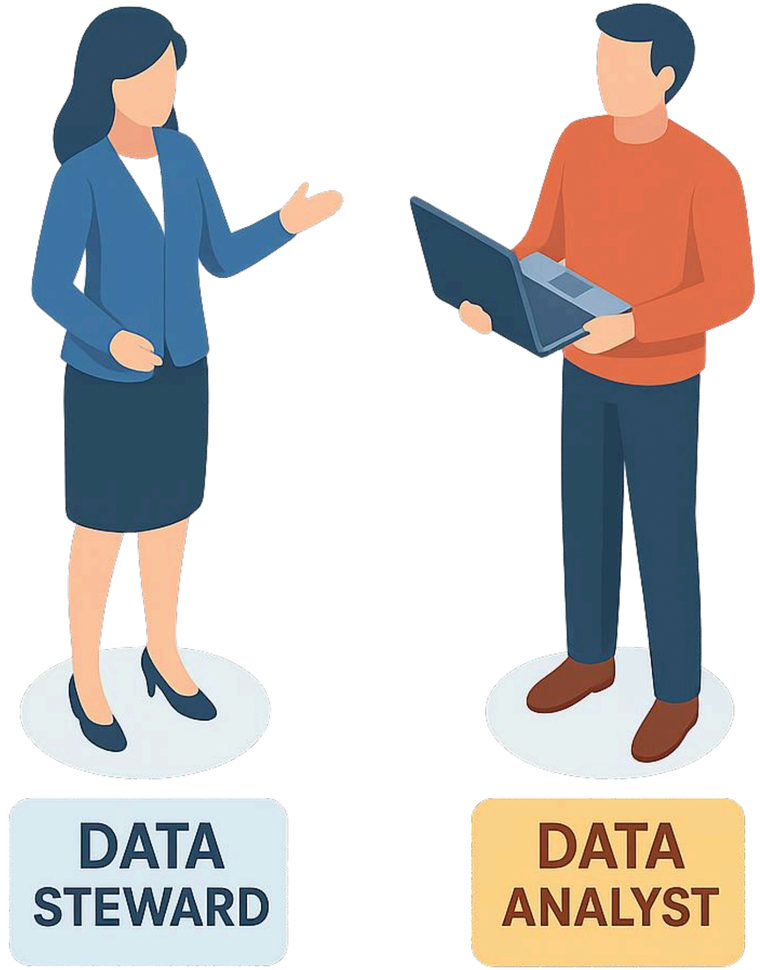
Responsibilities:

- Ensures data correctness, completeness, and reliability.
- Collaborates with data owners and business users to fix data issues.
- Defines standards for data formatting, normalization, and enrichment.

Data Analyst

Responsibilities:

- Analyzes and interprets data to support business decisions.
- Develops reports, dashboards, and KPI metrics.
- Uses BI tools for data visualization and insights.



Technical roles

Data Engineer

Responsibilities:

- Designs and maintains ETL/ELT processes.
- Integrates data sources into a DWH or Data Lake.
- Optimizes storage and processing of large datasets.

Machine Learning Engineer (ML Engineer)

Responsibilities:

- Develops and deploys machine learning models.
- Works with ML Ops and automates model training pipelines.
- Integrates models into business processes.

Data Architect

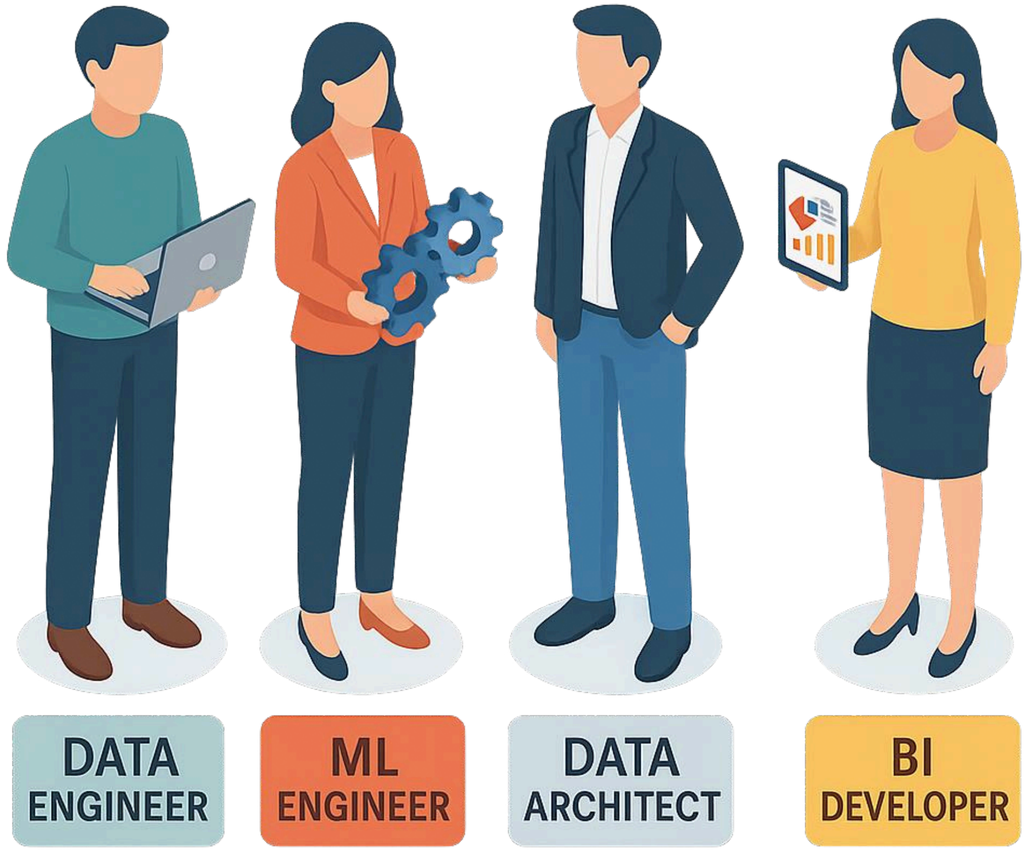
Responsibilities:

- Designs data storage architecture (DWH, Data Lake, Data Mesh).
- Creates integration schemas between systems
- Defines standards for data storage and processing.

BI Developer

Responsibilities:

- Builds analytical data marts and BI dashboards.
- Optimizes SQL queries and data models.
- Works with OLAP cubes and BI tools.



Supporting roles

Data Governance Manager

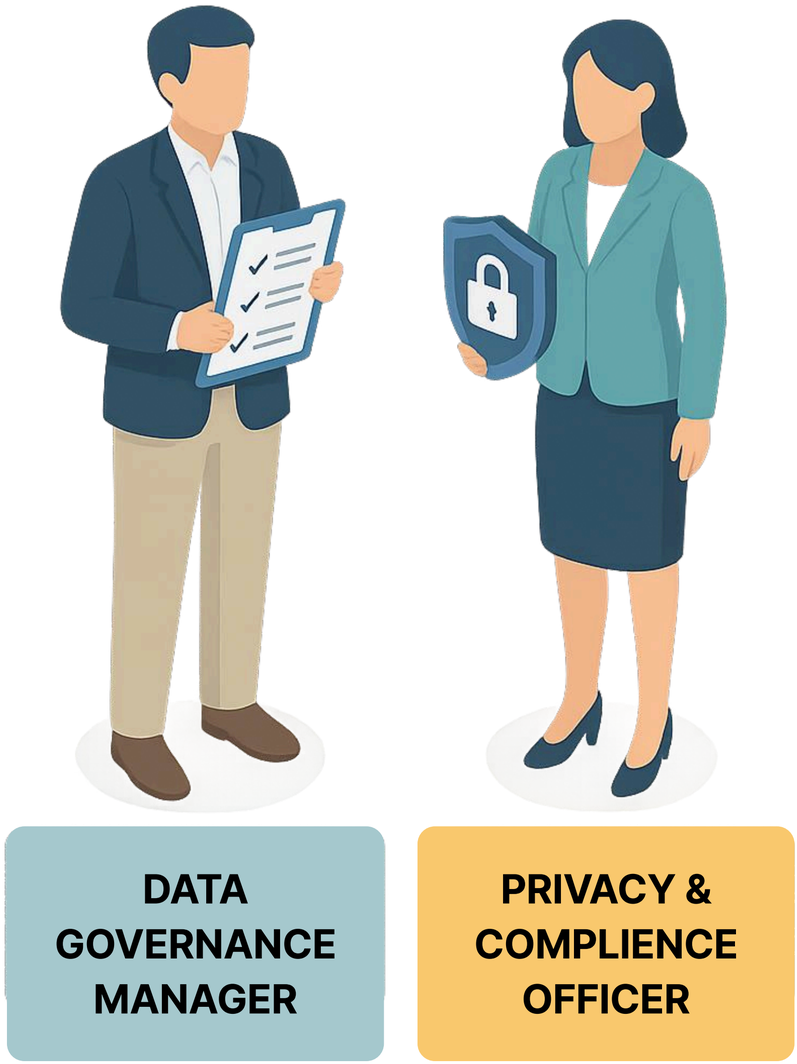
Responsibilities:

- Implements data governance policies across the organization.
- Monitors compliance with standards and regulatory requirements.
- Collaborates with data owners and stewards.

Privacy & Compliance Officer

Responsibilities:

- Ensures data compliance with laws and regulations (e.g., GDPR, HIPAA).
- Develops policies for storing and processing personal data.
- Manages access to sensitive and confidential information.



Data products

When we get a product in a boiling ocean of data?

Data products

Definition

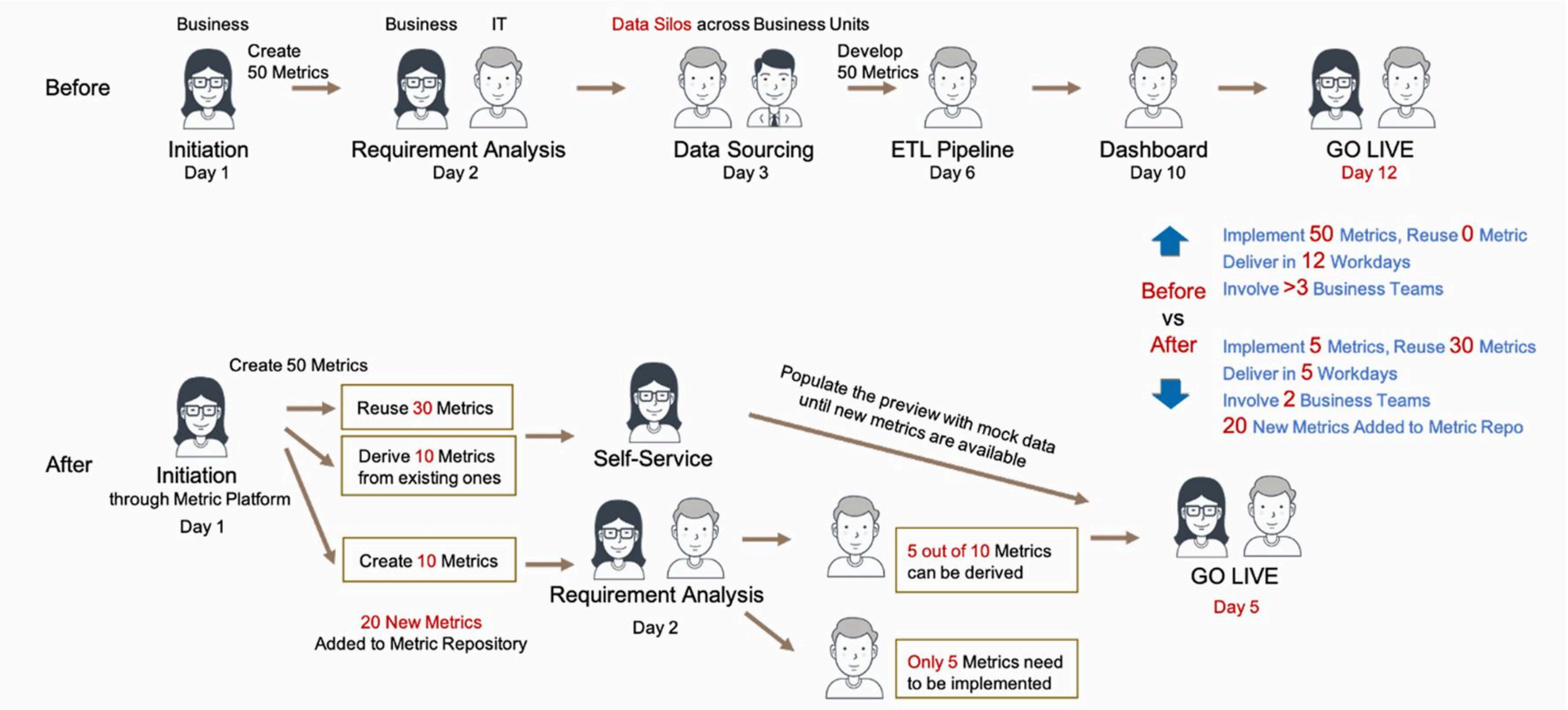
What is a data product?

A **data product** is a managed set of data, presented in a user-friendly form for use within the organization or by external users. A data product delivers high-quality, accessible, and valuable data that can be used in analytics, BI, machine learning, and automated processes.

Data products are a key element of the **domain-oriented** approach, where each business domain manages its data as a product.

Data products

Reuse of data

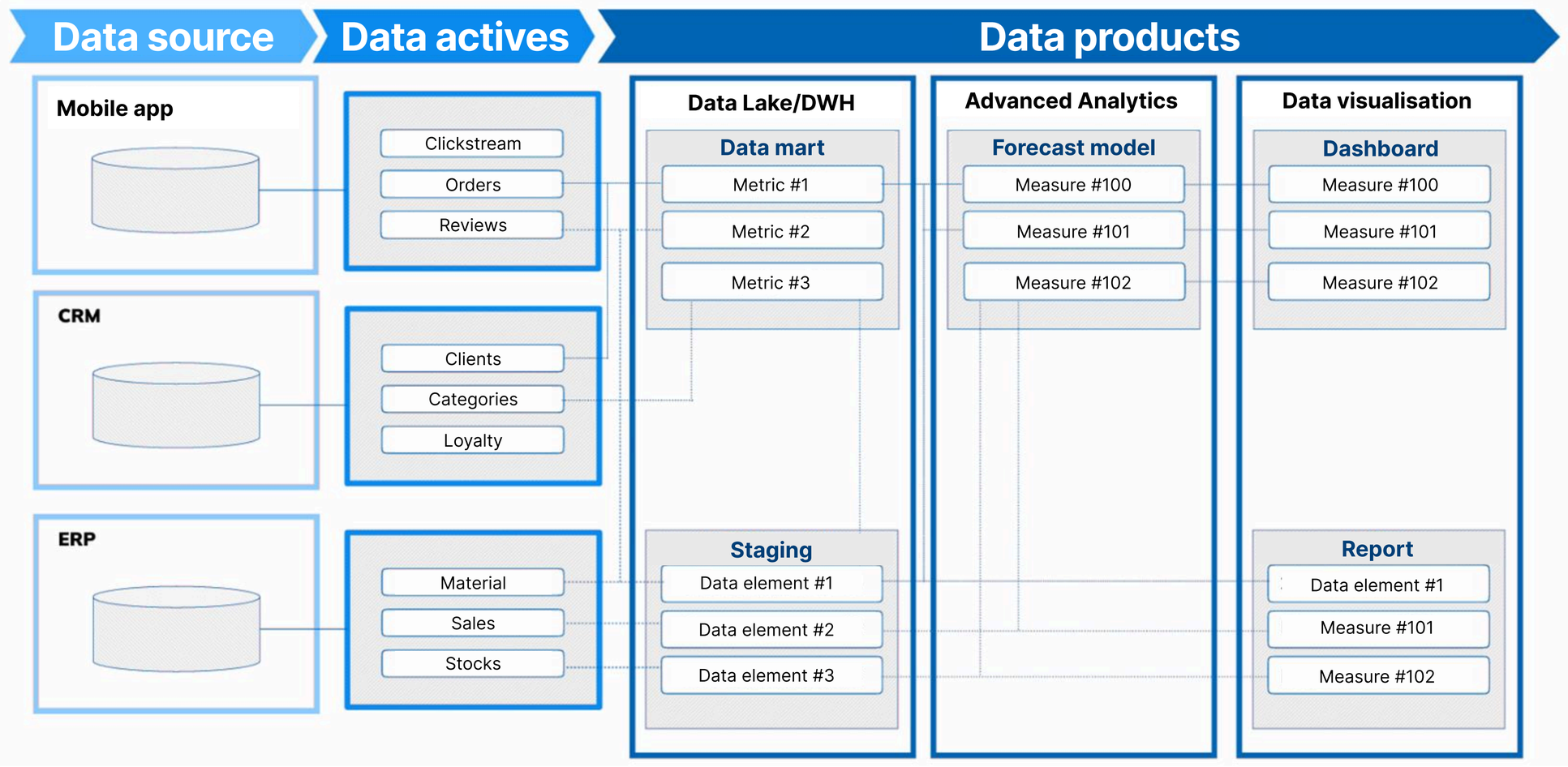


Data products

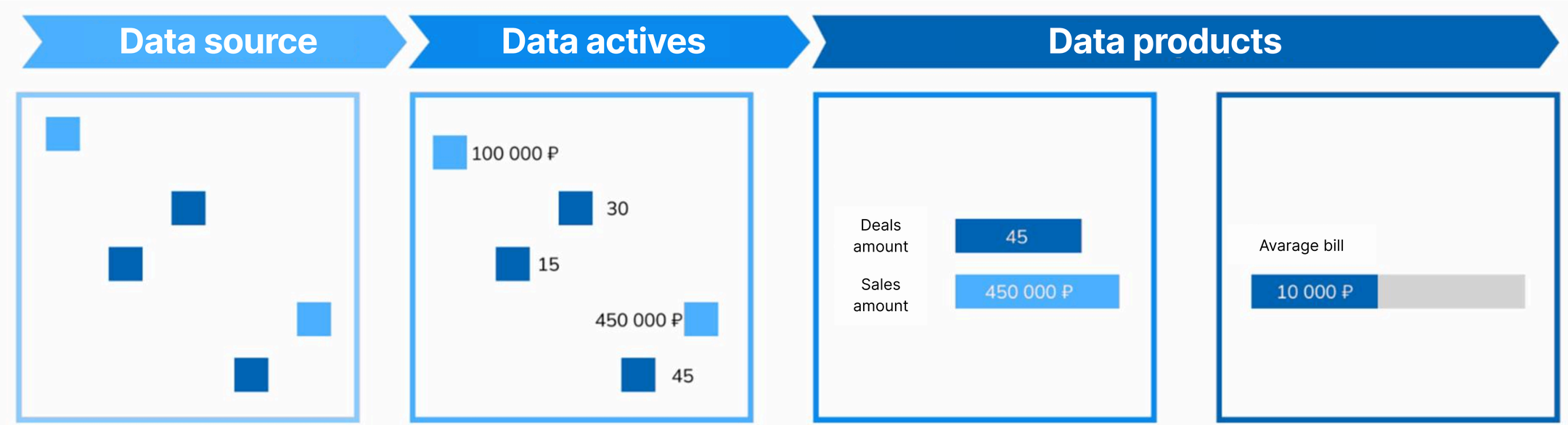
Classification

	Data Product	Data Storage	Analytics Platform	Content	API	
Data Visualization	Management Dashboards	-	BI	Report	-	Interactive visualization
	Reports	-	BI	Report	-	Text-based data view
Analytics	Prescriptive Analytics	-	Simulators	-	JSON	What actions should be taken?
	Predictive Analytics	-	ML Models	-	JSON	What might happen?
	Diagnostic Analytics	-	Statistical Models	-	JSON	Why did it happen and is happening?
	Descriptive Analytics	Cubes	Predefined Queries	-	JSON	What happened?
Data Operations	Aggregated Data	Data Marts	Sandboxed Data	Dataset	JSON	Customer and product data
	Preliminary Calculations	Dynamic Views	Sandboxed Data	Dataset	JSON	Calculated fields and metrics
	Consolidated Data	Core Data Layer Tables	Sandboxed Data	Dataset	JSON	Data from various sources
	Standardized Data	Staging Tables	Sandboxed Data	Dataset	JSON	Raw data formatted to business standards
	Raw Data	ODS Tables	Sandboxed Data	Dataset	JSON	Unmodified copies of data assets

Data value chain



Delivery of a data product



Facts

Events that occur during business processes are recorded in transactional systems as documented facts.

Data

Data in transactional systems represent matched facts — actual and quantitative parameters.

Metrics

A metric reflects actual and quantitative parameters as a cumulative result.

Indicators

An indicator describes quantitative characteristics of a process, but unlike a metric, it has a target or normative value.

Data Pipeline

How to transfer data from the source to the target?

Data Pipeline

Definition

What is a Data Pipeline?

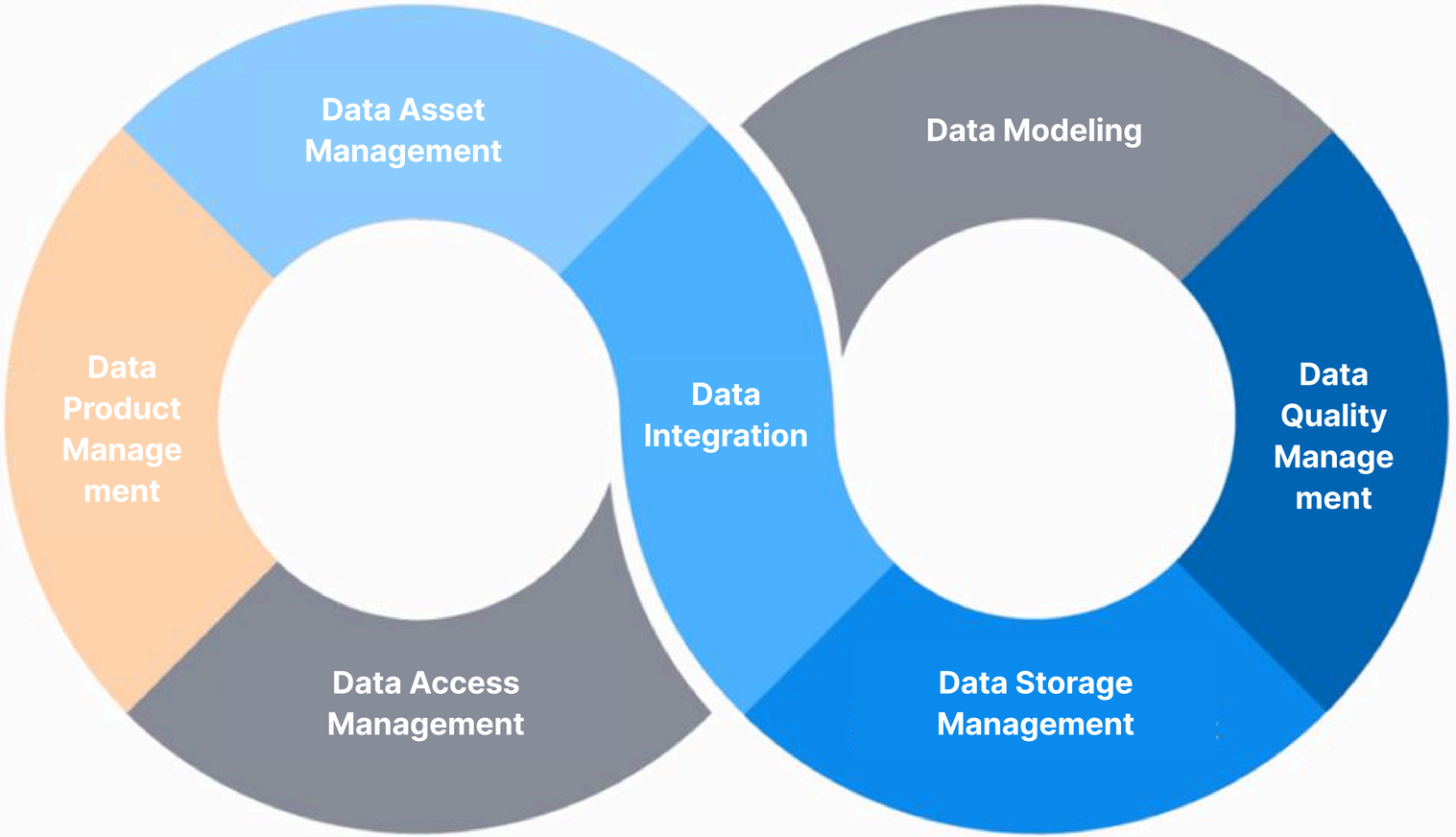
Represents a **highly efficient system** for developing and maintaining data products.

Goal of a data pipeline is to increase the productivity of the data office team when performing data-related operations

Primary objective of the pipeline is to organize workflows in a way that the **execution time** of data operations **is deterministic**.

Data Pipeline

Data-ops



DataOps represents an end-to-end data pipeline that ensures data is delivered reliably, efficiently, and with high quality.

It covers the full lifecycle — from data integration and storage to modeling and quality management — while also enabling access, governance, and product-oriented use of data.

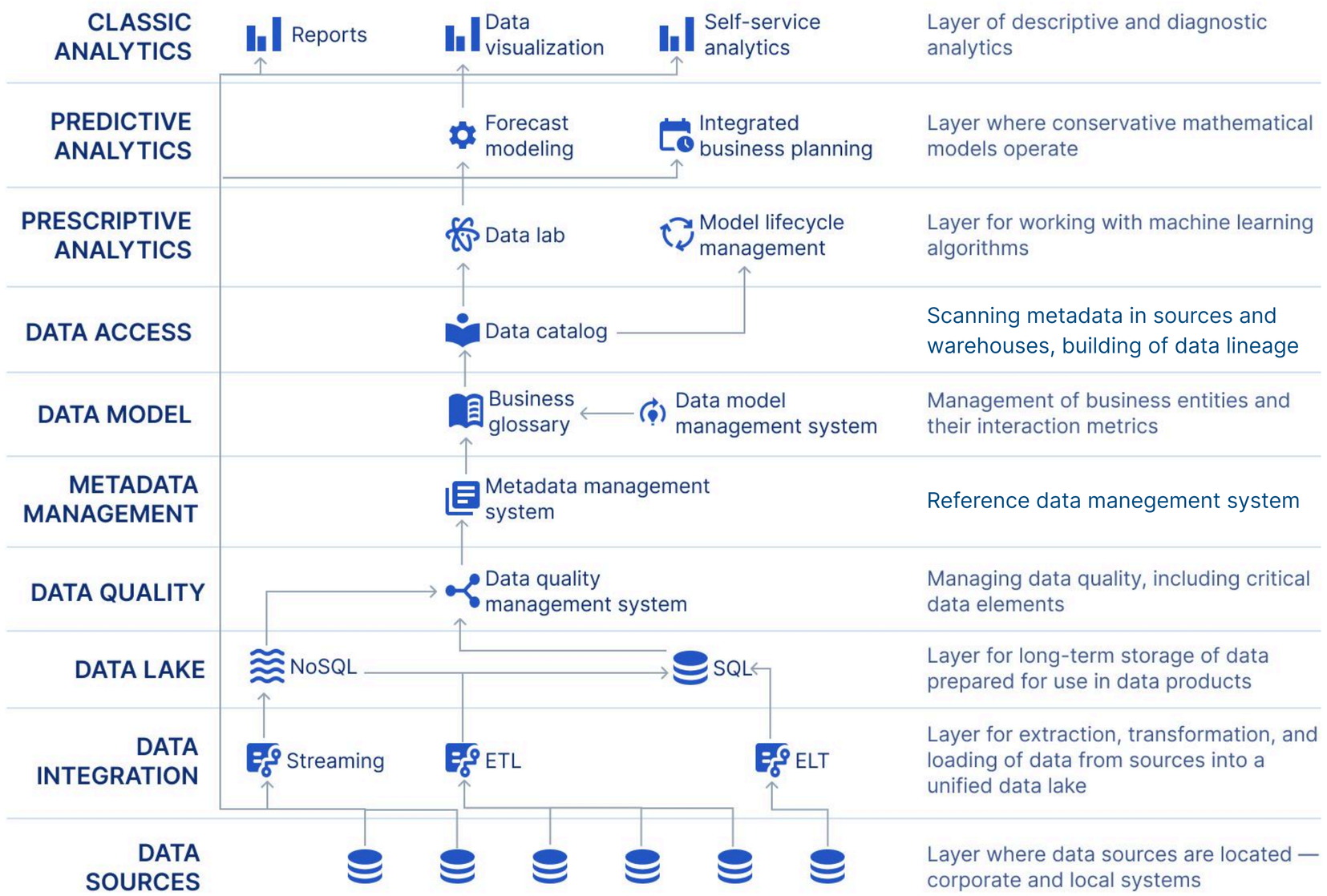
On the left, the focus is on data enablement and usage (data products, access, and asset management).

On the right, the focus is on data foundation and control (modeling, storage, and quality). Together, these components create a continuous, governed data flow that supports scalable analytics and faster business decision-making.

Data Pipeline

Components

One of the most common ways to organize a data pipeline.



Metadata, architecture and data modeling

Everything you need to know about your data

Definition

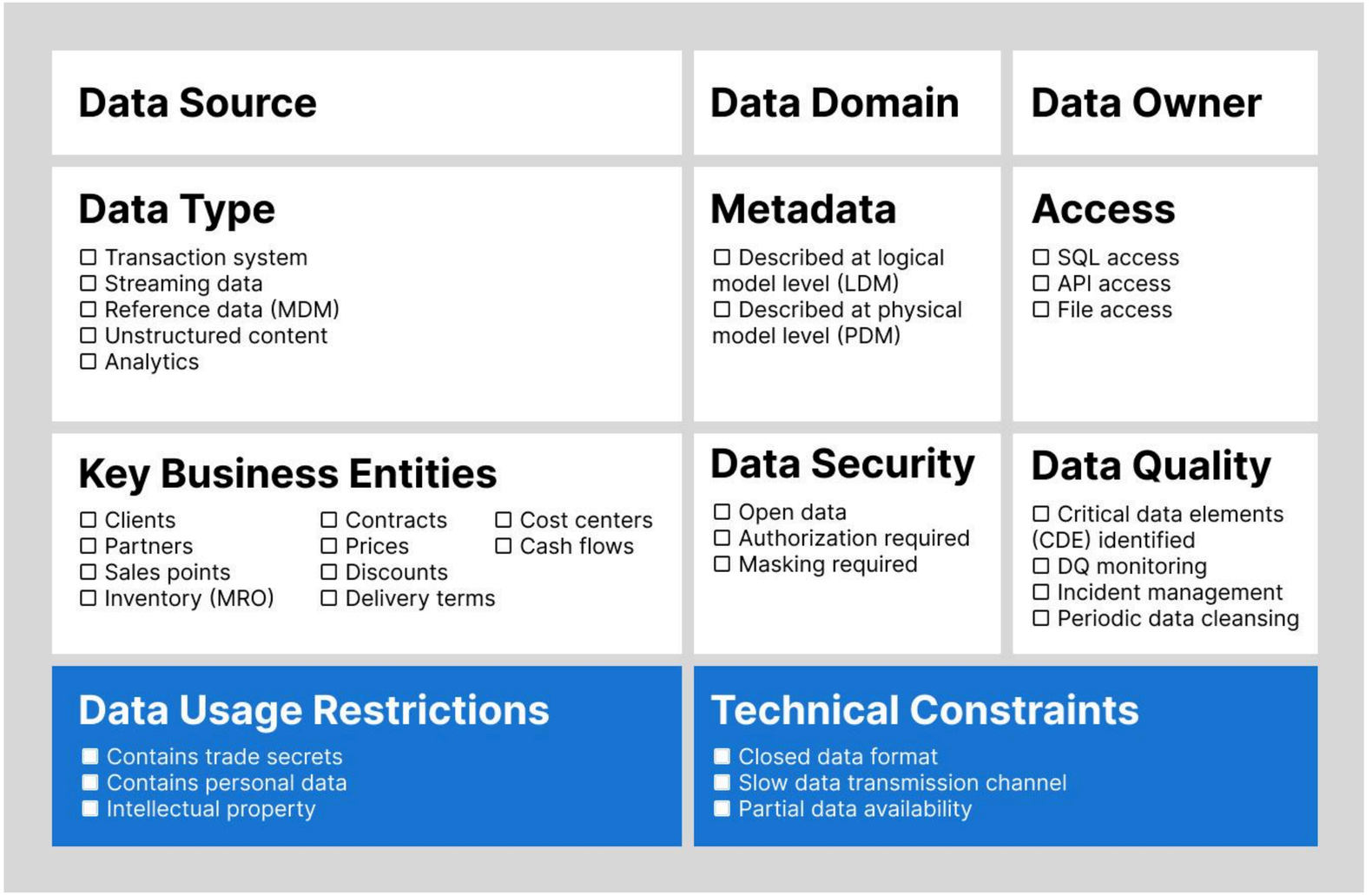
What is a Business Glossary and Data Catalog?

Business Glossary and **Data Catalog** are key tools of Data Governance, ensuring a shared understanding of data across the organization and enabling control over its usage.

The Business Glossary answers the question:
“**What do the data mean?**”

The Data Catalog answers the question:
“**Where is the data stored and how can it be found?**”

Data Asset Management



Definition

What is a Data model?

A data model is a formal representation of the **structure of data**, their relationships, and the rules for storing and processing them. It defines how **data is organized**, **connected** and **used** within information systems.

Conceptual Model – a high-level description of the subject area, without technical details.

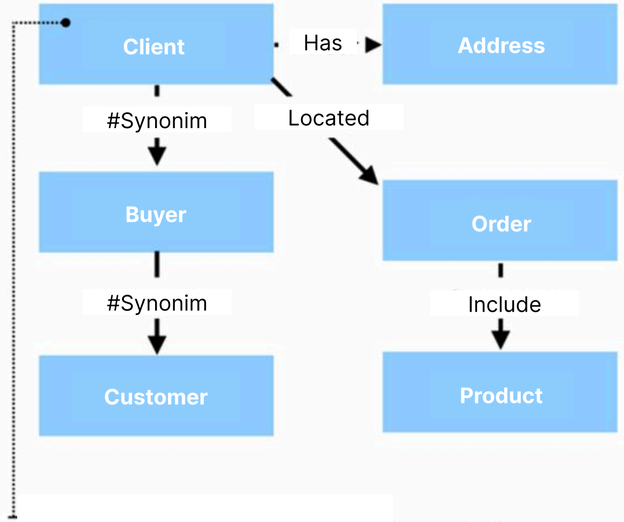
Logical Model – a detailed description of entities, attributes, and relationships between data.

Physical Model – the implementation of the data structure in a specific database (e.g., SQL tables, indexes, keys).

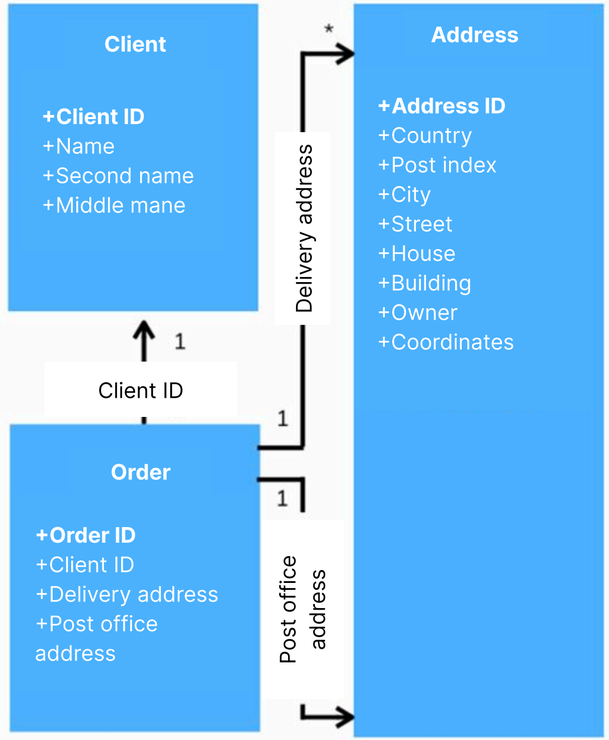
Metadata, architecture and data modeling

Data Models

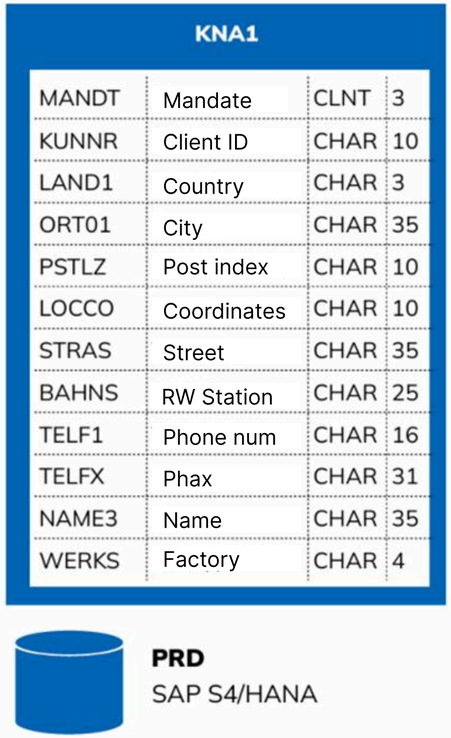
Conceptual Data Model



Logical Data Model



Physical Data Model

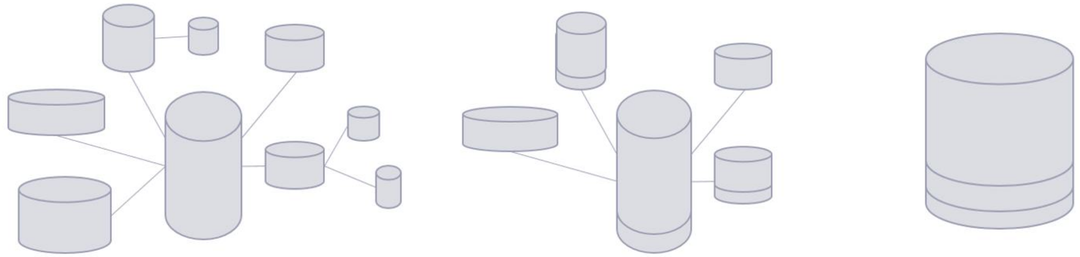


Data Warehouse Models

Model		Approach	Structure	Strengths	Best for
Kimball (Dimensional)		Bottom-up	Fact & Dimension tables (Star/Snowflake schema)	Simple, fast for analytics and BI	BI, dashboards, self-service analytics
Inmon (Corporate Information Factory)		Top-down	Normalized 3NF EDW + dimensional Data Marts	Enterprise consistency and long-term governance	Enterprise-wide consistent data management
Data Vault		Hub-and-spoke	Hubs, Links, Satellites	Scalable, audit-friendly, history-tracking	Complex, fast-changing environments with multiple sources
Anchor Modeling		Flexible normalization	Anchors, Attributes, Ties	High adaptability to changes	Environments with evolving requirements
Ensemble Modeling		Meta-framework	Depends on chosen ensemble (e.g., Vault or Anchor)	Combines benefits of various models	Teams choosing best-of-breed patterns
Hybrid Model		Combination	Mix of Kimball, Inmon, or Vault	Tailored to business and technical needs	Organizations with hybrid data needs

Types of physical Data Models from BI perspective

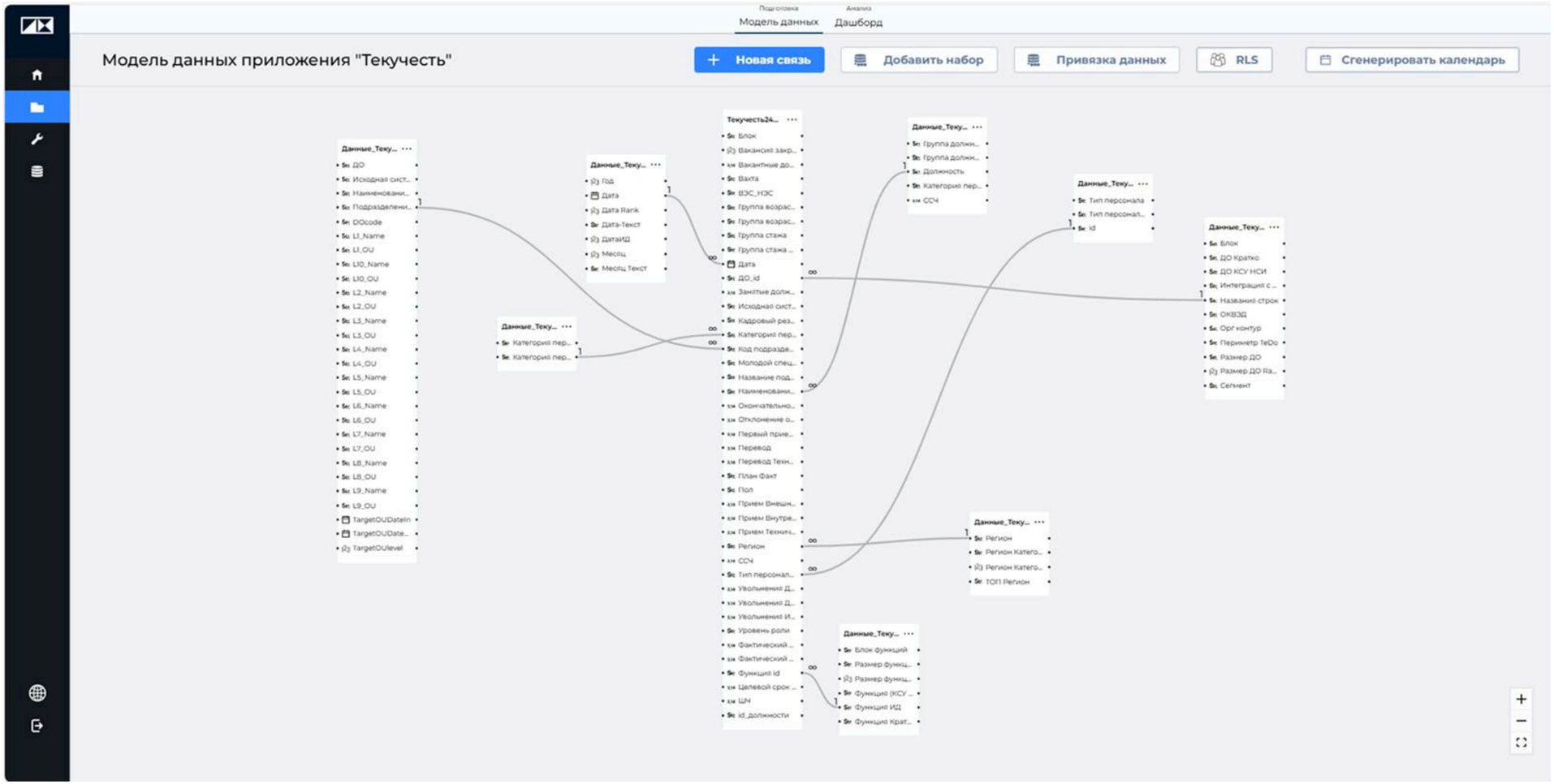
The **fewer the links** between tables, the **faster** recalculations happen in the application when **filtering data**.



Metric	Snowflake	Star	Flat Table
Response time	Satisfactory (🔴)	Good (🟡)	Excellent (🟢)
RAM usage	Normal (🟡)	Good (🟢)	Poor (🔴)
Script complexity	Low (🟢)	Medium (🟡)	High (🔴)
Expression complexity in UI	Low (🟢)	Low (🟢)	High (🔴)

Metadata, architecture and data modeling

Example of Data Model in PIX BI



Definition

What is a Data Lineage?

The process of tracking the flow of data from its source to its final usage. It shows how, where, and by whom the data has been modified, helping to ensure **transparency**, **control**, and **data quality**

Data transparency – employees understand where the data comes from and how trustworthy it is.

Simplified auditing and incident investigation – easier to trace data issues.

ETL/ELT process optimization – better control over data transformations.

Data quality

Can I trust you?

Data quality

Definition

What is a Data quality?

The degree to which data is **accurate**, **complete**, **reliable**, **relevant**, and **consistently structured** for its intended use.

High-quality data ensures that decisions, operations, and analyses based on it are **trustworthy** and **effective**. It is often assessed through dimensions like accuracy, completeness, consistency, timeliness, validity, and uniqueness.

Data quality

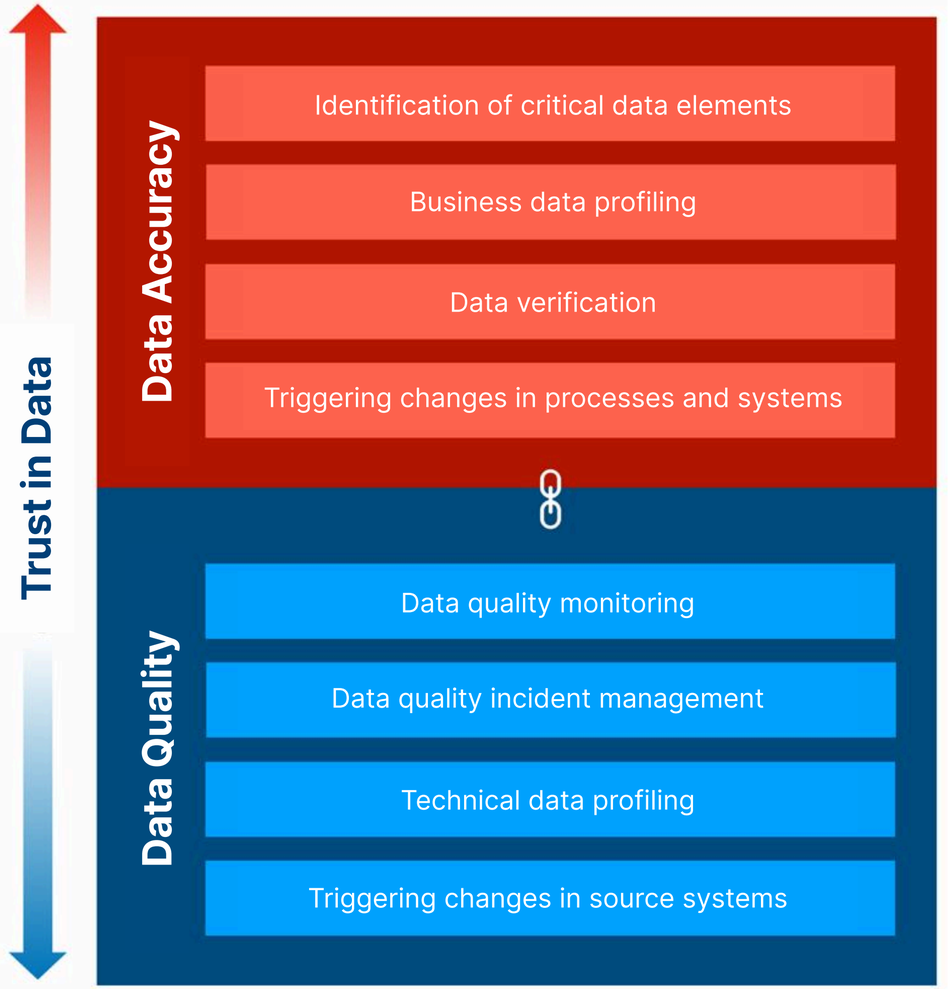
Trust for Data

Data Accuracy: the alignment of data with reality
Data accuracy defines how well data reflects the real state of things within a company.
Accuracy cannot be established solely by assessing data quality.
(Example: a customer's address may match all quality checks in the national address register, but if the address is outdated, a shipment may fail and cause losses.)
Ensuring data accuracy requires the involvement of data owners.
It also assumes the following roles are assigned:

- Data domain administrator
- Data accuracy analyst

Data Quality: compliance of data with requirements
Data quality measures how well a set of data characteristics meets declared requirements.
Data quality can be quantified using a DQI (Data Quality Index).
Data quality management is a coordinated activity involving control and governance over the organization’s data quality.
This includes assigning the following roles:

- Data quality manager
- Data quality engineer



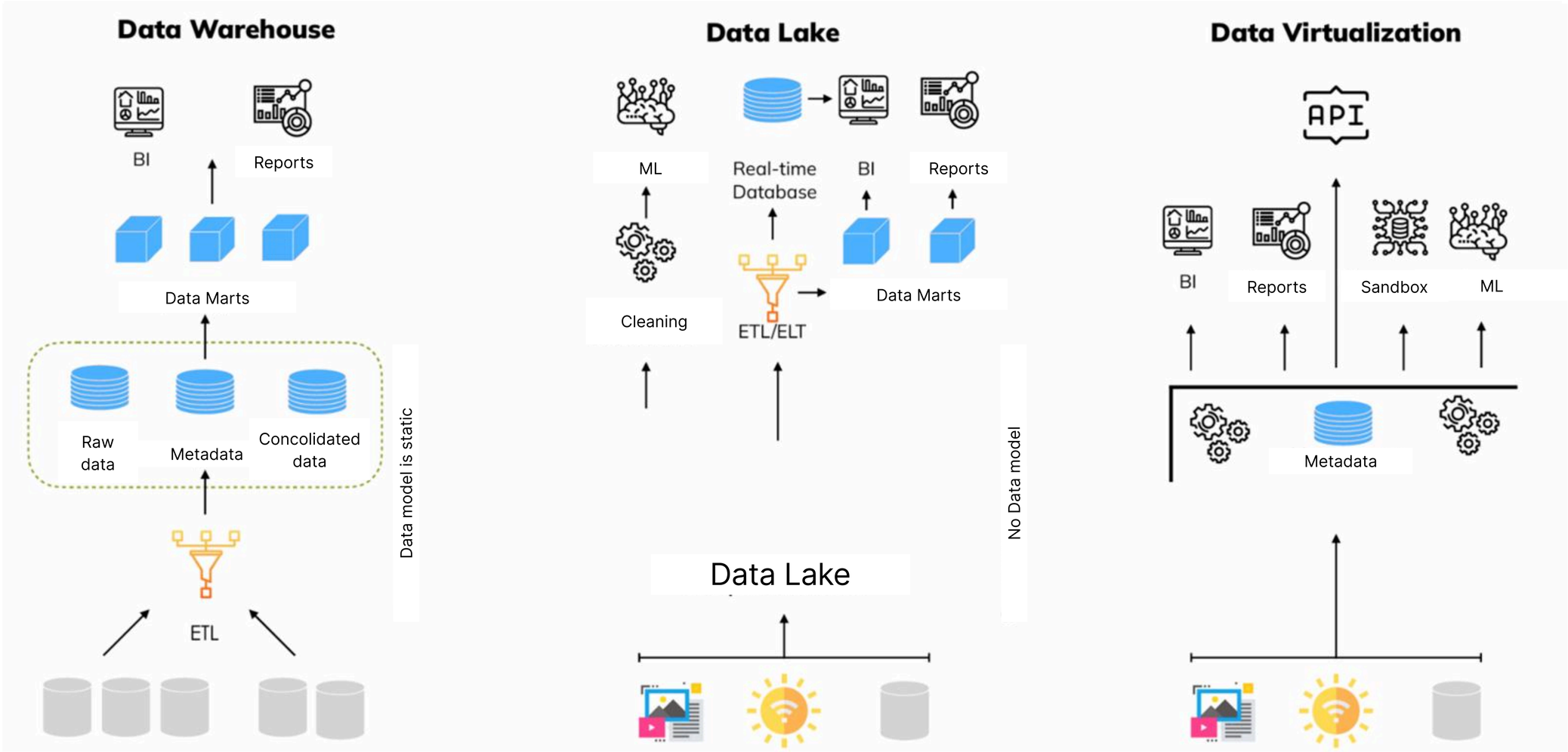
Data quality criteria

Criterion	Description	Example
Format (Validity)	- Data type description - Range of acceptable values or format	- 10-character string - First 4 characters are digits, next 6 – digits and Latin letters - Empty values are allowed
Accuracy	- Precision of measurement/calculation of each data attribute	- The number of days from the account opening date is calculated correctly
Completeness	- Full list of required elements is reflected in the system	- When transferring data from the source to the warehouse, each transaction was successfully transferred
Timeliness	- Requirements for the data extraction time interval - Defined update time boundaries	- Data from the beginning of the current year is required by 12:00 - Actual data should be in the warehouse by 8:00
Uniqueness	- Whether different data elements can have the same value for a given attribute	- Each credit record in the portfolio must have a unique loan contract
Consistency	- Presence of specific business rules for verifying data accuracy	- End-of-day balance for an account = total of all transactions + opening balance for the day

Storage and integration

We need a lot of space

Storage architecture



Layers of data

Hot Data

In-memory data storage layer	Transaction processing Storing critical data for real-time access Real-time analytics	Hybrid SQL/NoSQL architecture

Warm Data

Data stored on fast SSDs	Data stored on fast SSDs Analytical operations: aggregation, filtering, sorting MPP data processing Deep data analysis Long-term data storage	• Columnar databases • Relational databases • Graph databases

Cold Data

Data stored on cheaper, high-capacity disk drives	Data stored on cheaper, high-capacity disk drives Storage and processing of large data volumes (petabytes) Storage of files, binary objects, and unstructured content	• HDFS/Hive • S3

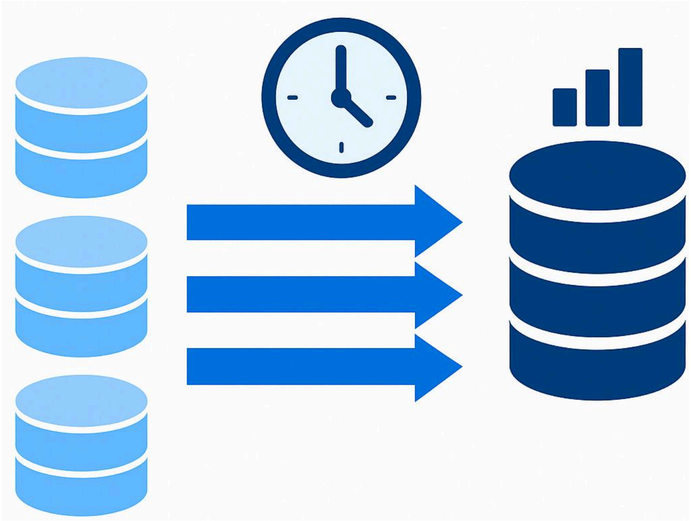
Definition

What is a Data integration?

The process of combining data **from various source systems** into a unified data repository (such as a Data Warehouse or Data Lake) to provide a **consistent**, **complete**, and **accessible** view of information.

The **goal** of data integration is to make data from different systems (CRM, ERP, finance, operations, etc.) **available in a single format** for analytics, reporting, and decision-making

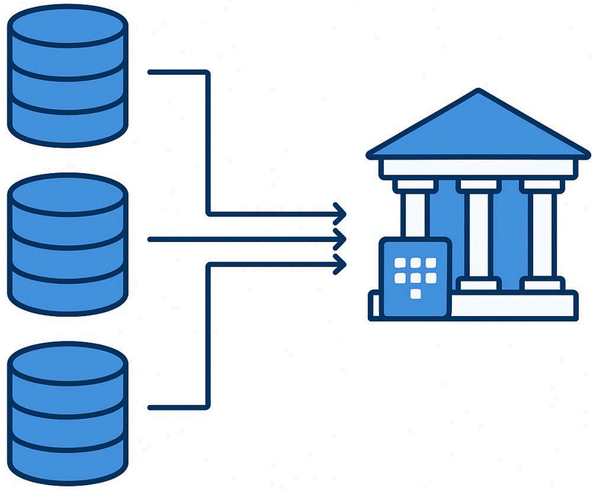
Types of integration



Batch Integration (ETL/ELT)

- Periodic data extraction (e.g., daily, hourly)
- Suitable for large volumes with less frequent updates
- Uses staging areas, transformation logic.

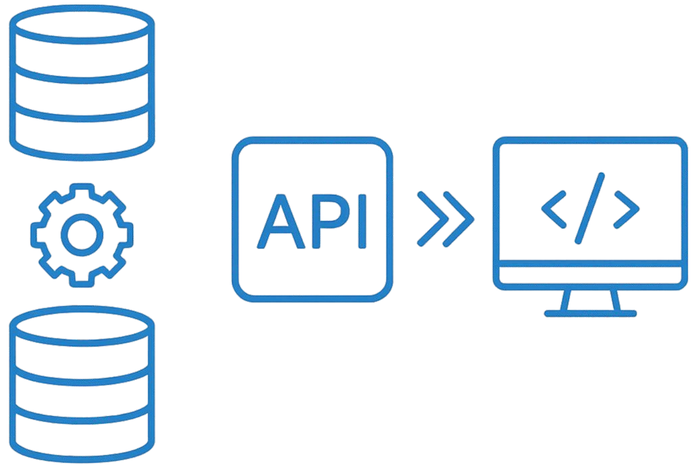
✅ Pros: Simple, controlled
⚠️ Cons: Delay in data freshness



Real-Time Streaming (CDC, Event-Based)

- Change Data Capture (CDC) tracks and pushes changes in real time
- Ideal for high-frequency transactional systems

✅ Pros: Near real-time updates
⚠️ Cons: Higher complexity, requires robust monitoring



API-Based Integration

- Pulling data via REST/GraphQL APIs from cloud or SaaS apps
- Good for modern, service-based systems (CRM, ERP, etc.)

✅ Pros: Flexibility, direct access
⚠️ Cons: Rate limits, error handling required

Master data management

Master of data

Master data management

Definition

What is Master Data?

Unified reference books, classifiers, regulatory, and reference data used across different systems and processes within a company.

Core Principle: "One version of the truth" – standardized, consistent data shared across all departments.

Key Objectives of a Corporate MDM System

Centralized storage

Reference and regulatory information (reference books, classifiers, regulatory data)

Data quality assurance

Validation rules and master data

Change management

Version control and approval workflows

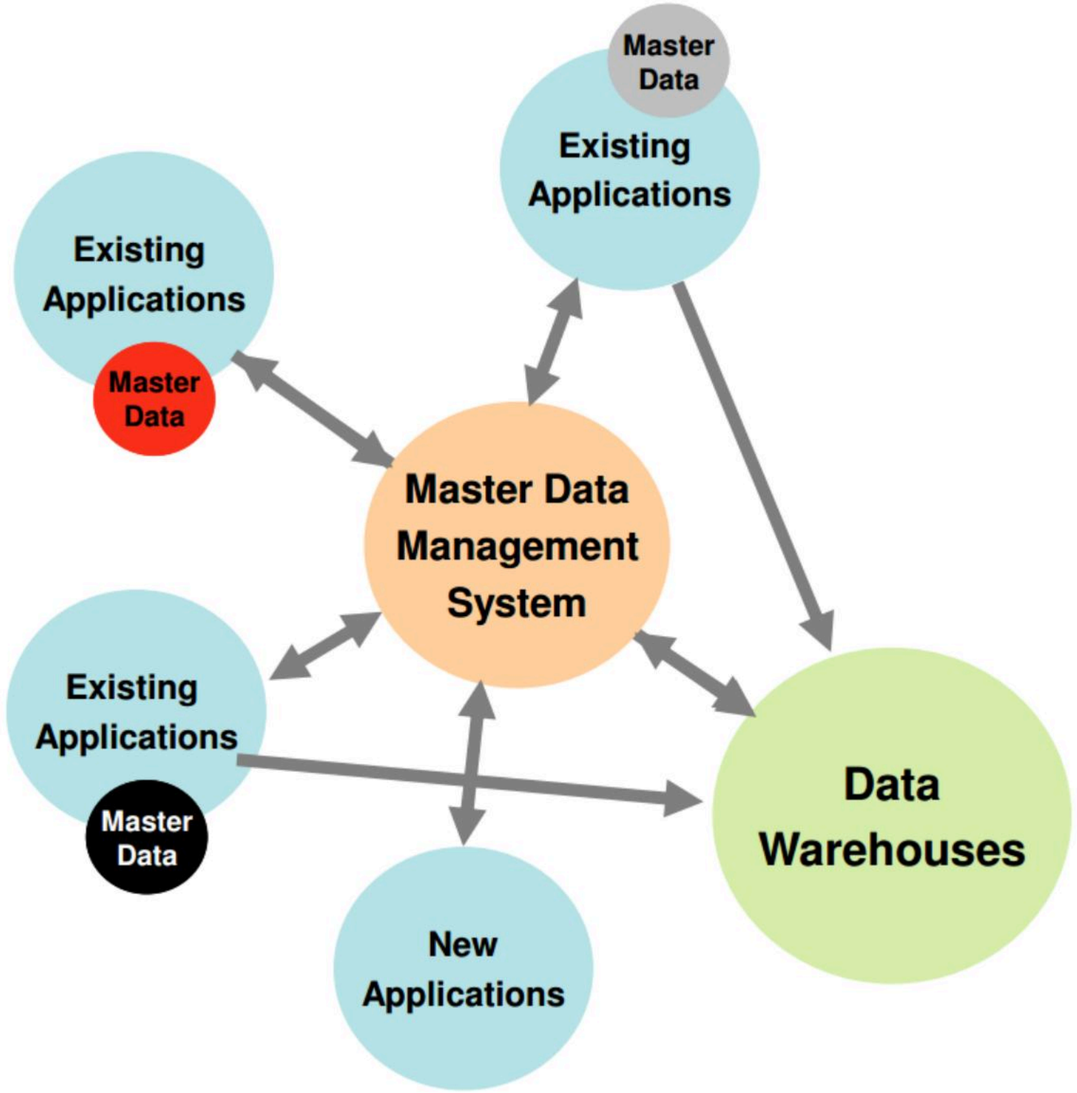
Integration

With ERP, CRM, DWH, and other systems

Master data management

MDM

An MDM system (Master Data Management system) provides a unified view of all business components, including data sources, ownership, quality, completeness, and potential data usage



Types of corporate data

Unstructured data

Text, emails, and other data that do not have a formally defined or described structure.

Structured (transactional) data

Data with a formally defined schema

Metadata

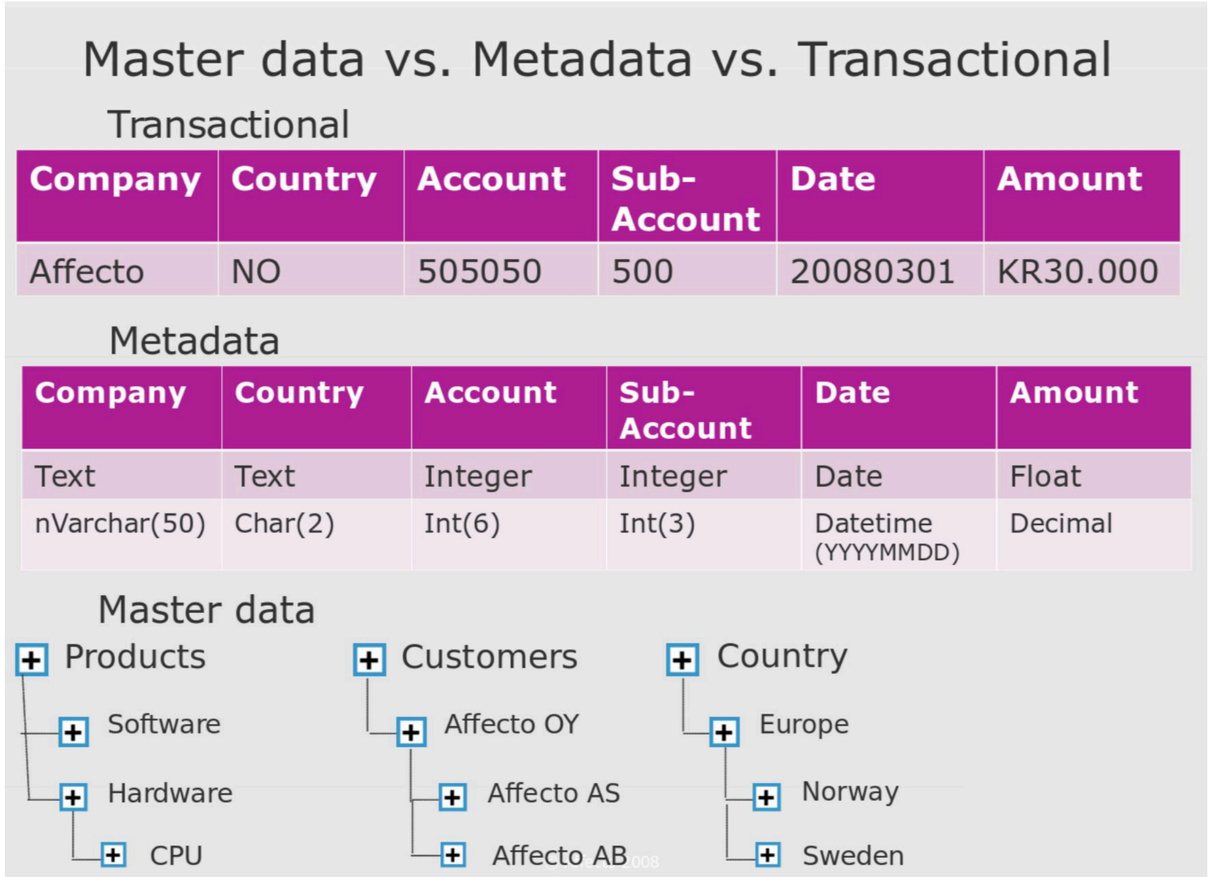
Data that describes other data, such as a customer database schema, configuration file, or report template.

Master data

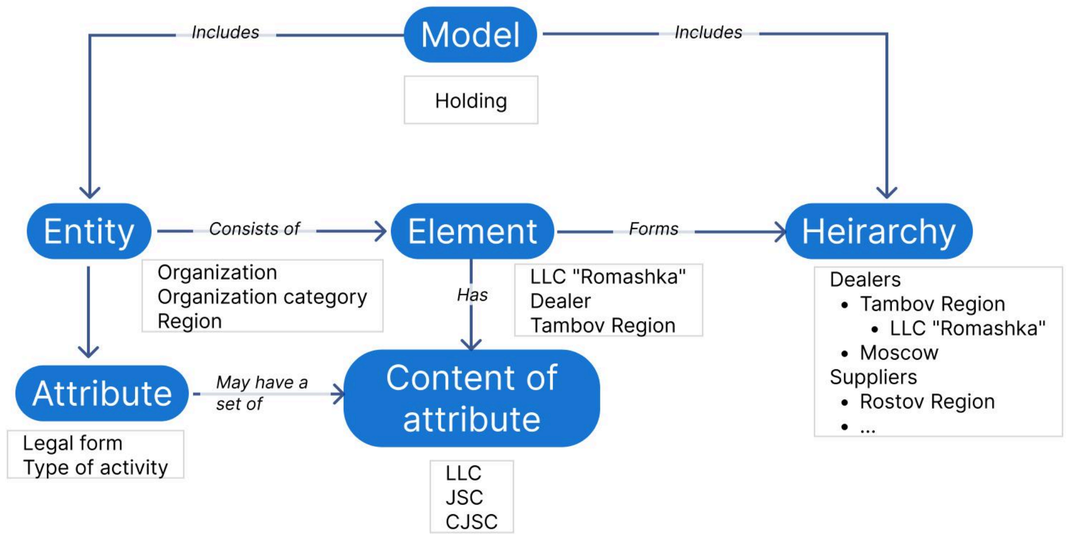
Data containing key business information, including details about customers, products, employees, technologies, and materials.

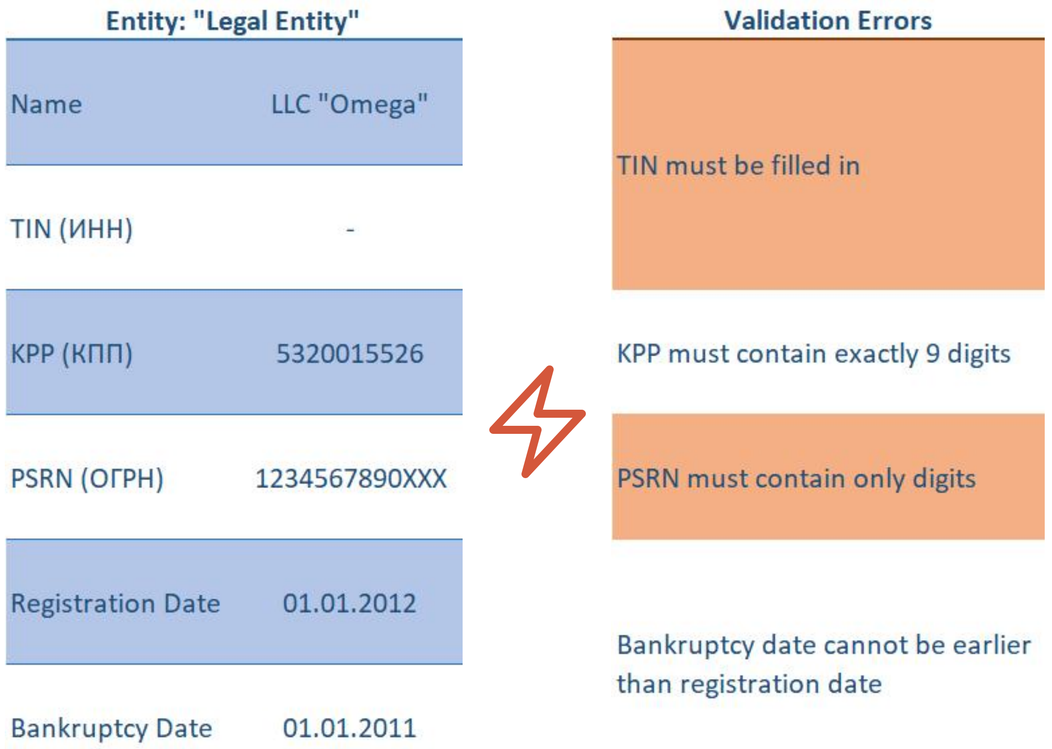
Transactional data

Transactional data refers to the information that is generated by and records the details of business events or transactions. These are usually dynamic, time-stamped records of activities that change over time and are crucial for business operations.



Master data management





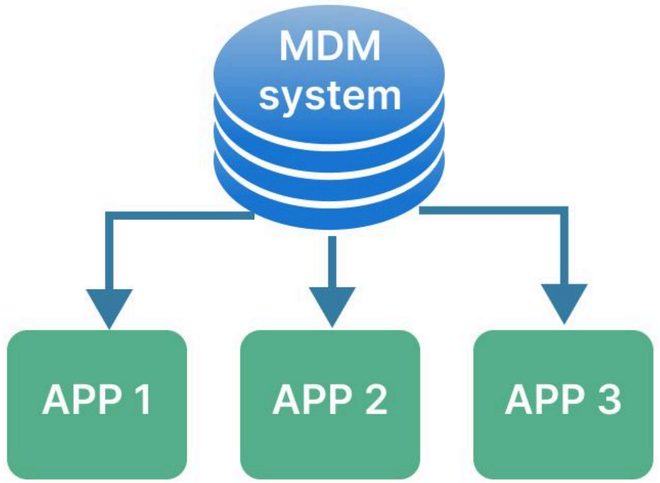
Structure of master data

Every master data record typically includes a **unique identifier**, to ensure consistency across systems, as well as **relationships between entities**, such as the link between a customer and their sales representative, or a product and its supplier. **Hierarchies** are used to organize data into logical structures — for instance, a product hierarchy may consist of category, subcategory, and item.

Rules of master data validation

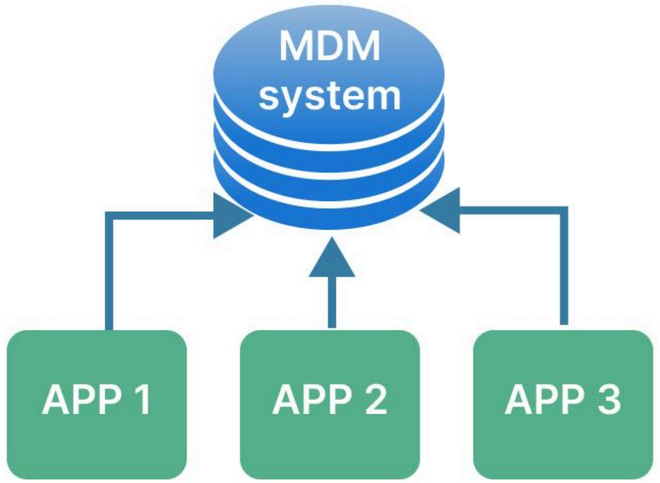
Master data also includes **validation rules** that define acceptable values (e.g., mandatory fields, format checks)

Master data management



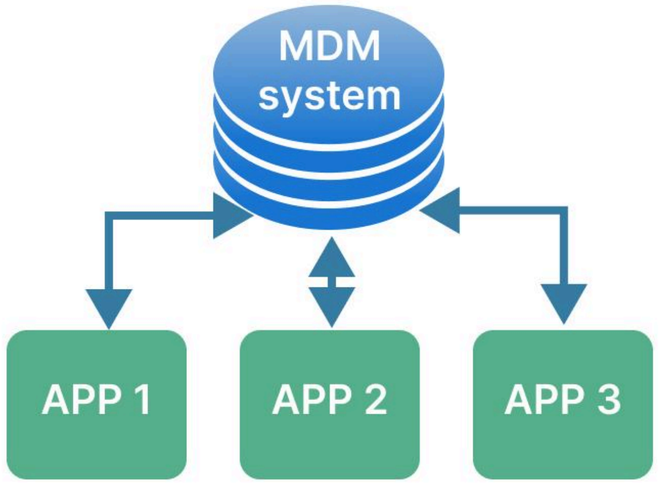
Centralised

The reference data in MDM system is considered the golden source: it is maintained there and **distributed to other systems**. At the same time, the creation and editing of reference data in other IT systems is **prohibited**



Analytical

All master data elements are **created in the client systems** and then sent to the MDM system, where these elements are used to form a reference data record.



Harmonised

Allows data to be entered in IT systems and then **matched with existing records**, d**etects potential duplicates**, **resolves conflicts** arising from simultaneous changes to the same data in different systems, and synchronizes master data across IT systems.

Business intelligence and advanced analysis

Cherry on your cake

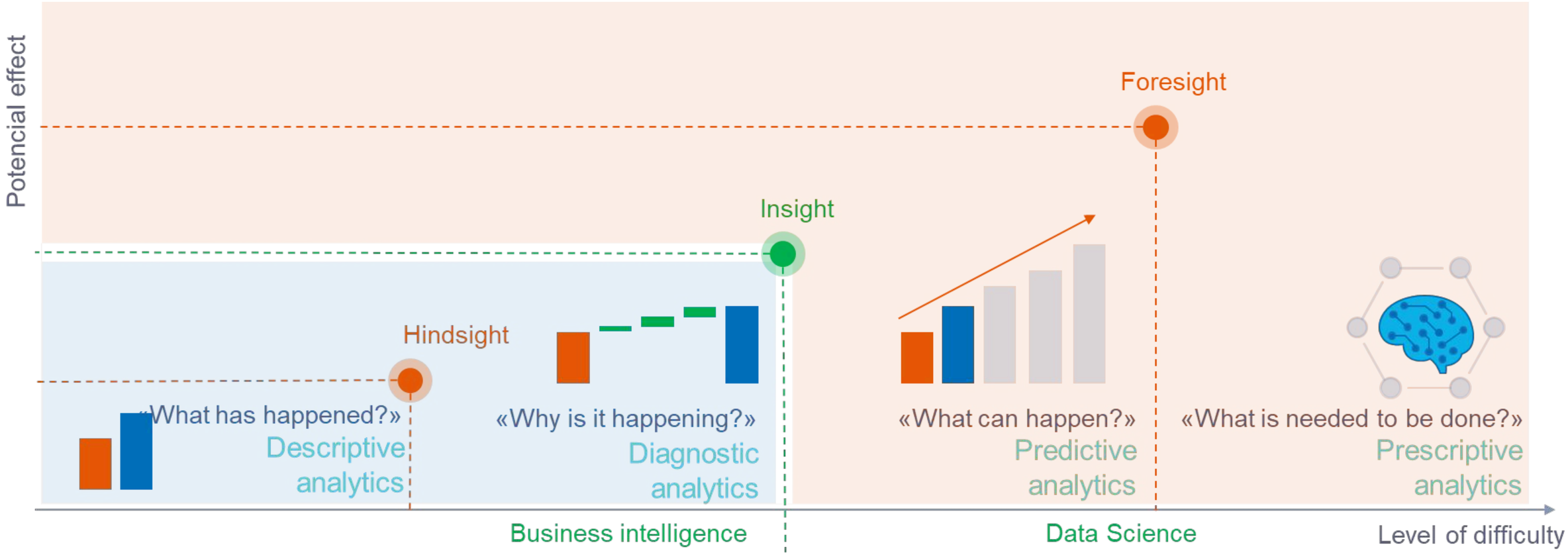
Definition

What is Business intelligence and advanced analysis?

Business Intelligence refers to technologies and processes used to collect, integrate, analyze, and visualize business data to support decision-making.
It focuses on historical and current data, typically through dashboards, reports, and KPIs.

Advanced Analytics involves the use of sophisticated techniques like machine learning, predictive modeling, and statistical algorithms to forecast future outcomes and discover patterns.
Unlike BI, it is more forward-looking and often used for optimization and strategic planning.

Stages of analysis



BI approaches

Classic BI (IT Projects)

Development of analytical applications "turnkey" by a **development team** for business users as part of an IT project.

Self-Service BI (Data Lab)

Independent creation of applications by **business users** to solve daily tasks of data processing and visualization.

Business Requirements

Turnkey Analytical Applications

Team of Data Analytics
(BI developers, BI architects, designers, service team)

Data Request

Prepared data, tools, and knowledge

Business User

Independently developed applications

Business intelligence and advanced analysis

BI approaches

	Classic BI	Self-service Analytics
Tools	Luxms BI, PIX BI	PIX BI
Tasks	• Automation of regulatory reporting • Tasks with clearly defined and formalized business requirements	• On-demand analytics • Ad-hoc reporting • Business hypothesis testing
Data	• Data from Data Warehouse	• Data from Data Warehouse • User's local files
Implementation approach	• Project initiation required	• No need to initiate activity (for solving daily analytical tasks)
Developer involvement	• BI developer, BI designer, BI architect	• Independent application development by business users after training
Implementation time	• From 3 months to several years	• Depends on user's skill level, a simple solution can be created in a few days

Data Integration

Plus7Formit

Data Storage Management

Data Quality Management

Plus7Formit

Data Modelling

Enterprise Data Management

Plus7 EDM

Master Data Management

1C: MDM

Business Intelligence

Advanced Analytics